

First-Divergence Factorial Diffing for Post-Trained Language Models

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Abstract

Late-layer patching is a tempting one-cell summary of how post-training changes a model, but it can be unstable across prompt regimes. We show this at the first PT/IT next-token disagreement: on a factual/reasoning stress test, the late-only effect from PT upstream moves against the post-trained token (-1.18 logits), while the upstream $\times$ late-stack interaction remains positive ($+1.81$). We introduce first-divergence factorial diffing to measure that interaction directly: at the earliest shared-history prefix where a pretrained checkpoint (PT) and its post-trained descendant (IT) prefer different next tokens, we cross PT/IT upstream residual state with PT/IT downstream late stack and score the IT-vs-PT divergent-token margin. Across six dense PT/IT pairs, including Qwen2.5-32B, the interaction is positive in every family: the Dense-6 family-balanced estimate is $+2.44$ logits (95% CI $[+2.35, +2.52]$), and the Gemma-removed Dense-5 estimate is $+1.71$ $[+1.64, +1.78]$. The effect is not confined to immediate response openings: restricting to generated positions ≥ 3 still gives $+1.43$ logits ($[+1.30, +1.57]$), with all six family-specific intervals above zero. The sign is label-aligned under a five-family permutation null ($p=5 \times 10^{-5}$), and an OLMo-2 stage-lineage case study finds the largest adjacent-stage jump at Base $\rightarrow$ SFT, with smaller positive later-stage interactions. Supporting tests give a graded anatomy of the change: middle-positioned MLP substitutions transfer token identity more often, while late-positioned substitutions and native late MLP write-outs have larger margin/readout effects. The contribution is a reusable diagnostic protocol for paired-checkpoint diffing plus a cross-family characterization of how post-training changes next-token formation at the first PT/IT disagreement.

1. Introduction

Suppose a pretrained checkpoint (PT) and its post-trained descendant (IT) read the same prefix and arrive at different next-token preferences — say PT prefers A and IT prefers B . This is a model-diffing question: what changed inside the forward pass when the checkpoint became instruction-following? The practical risk is that a late-layer patch can be interpreted as a portable post-training mechanism when it is actually measuring compatibility between a late stack and the upstream residual state it expects. The point is not that late transformer blocks ordinarily ignore upstream state; they do not. The point is whether the PT $\rightarrow$ IT change is portable as a late-only update, or whether post-training makes late computation effective only after earlier layers have shaped the residual stream toward the IT candidate.

A layerwise signature made this counterfactual worth testing: IT checkpoints show delayed stabilization, meaning decoded intermediate layers stay farther from the same checkpoint’s final next-token distribution than PT layers do. Late-window graft/swap controls localize this delay to learned late MLP changes rather than generic late perturbations (§3.2). The first-divergence factorial asks the sharper model-diffing question: when PT and IT first disagree, does the learned IT late computation carry the IT-token margin by itself, or only in compatibility with an IT-shaped upstream state?

Definition 1: first-divergence factorial diffing. Let the first-divergence prefix be the earliest shared-history position where PT and IT prefer different next tokens; denote those tokens t_{PT} and t_{IT} . At a pre-specified late boundary, U_{PT} and U_{IT} are the residual states produced by running PT and IT on that identical prefix up to the boundary; L_{PT} and L_{IT} are the PT and IT downstream blocks from the boundary onward. The four hybrid forward passes are:

Upstream state	PT late stack L_{PT}	IT late stack L_{IT}
PT upstream U_{PT}	PT baseline	Late-only
IT upstream U_{IT}	Upstream-only	Matched IT

A readout R is specified separately (final norm, lm_head , real-token mask). The outcome is the divergent-token margin $Y_R(U, L) = \text{logit}(t_{\text{IT}}) - \text{logit}(t_{\text{PT}})$. The primary estimand is the upstream $\times$ late interaction:

$$[Y_R(U_{\text{IT}}, L_{\text{IT}}) - Y_R(U_{\text{IT}}, L_{\text{PT}})] - [Y_R(U_{\text{PT}}, L_{\text{IT}}) - Y_R(U_{\text{PT}}, L_{\text{PT}})].$$

A portable late-delta account of post-training would make this interaction small: the IT late stack would add a similar IT-token margin from PT-shaped and IT-shaped upstream states. We measure how far the paired PT/IT contrast departs from that

account. This also makes the design a validity diagnostic for simple late-only effects: if the one-cell late-only estimate and the four-cell interaction disagree, the late-only estimate is not a stable summary of the PT→IT change. Common-IT and common-PT readout variants (which score every cell with one fixed readout) and the full implementation are described in §2.2.

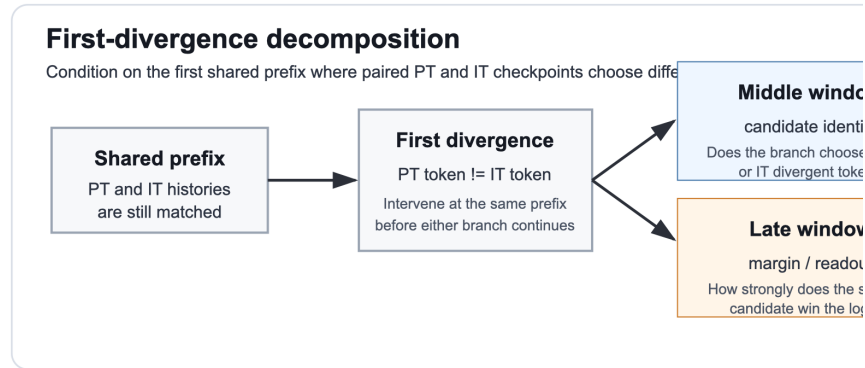


Figure 1: First-divergence factorial diffing schematic. The shared raw-token prefix reaches the first prefix where PT and IT prefer different next tokens (t_{PT} vs t_{IT}). At a pre-specified late boundary, upstream residual state (U_{PT} or U_{IT}) is crossed with downstream late stack (L_{PT} or L_{IT}) to produce four hybrid forward passes; the contrast across the four cells gives the upstream $\times$ late interaction on the divergent-token margin.

Result. Across six dense PT/IT pairs, the upstream $\times$ late interaction is positive in every family. The primary all-position estimate is $+2.44$ logits; a conservative later-position check at generated position ≥ 3 remains positive ($+1.43$) with all six family-specific intervals above zero. The interaction is label-aligned under a PT/IT label-swap null on the five-family raw-record subset ($p=5 \times 10^{-5}$) and remains positive even when the simple late-only term reverses sign on factual/reasoning prompts. Magnitude varies by family, prompt regime, and generated position; we treat that variation as regime structure of the PT→IT change, not as a failure of the claim. An OLMo-2 stage-lineage case study finds the largest adjacent-stage jump at Base→SFT, with smaller positive interactions at SFT→DPO and DPO→RLVR and a large Base→RLVR contrast. Details for Gemma, position thresholds, Qwen2.5-32B, and OLMo stages are reported in §3.1 and Appendices A/F.

Contributions. The paper has three connected contributions.

1. A validity diagnostic for late-only patching. The factual/reasoning stress test exposes the central failure mode: the simple PT-upstream late-only term can reverse sign (-1.18) while the factorial interaction remains positive ($+1.81$). One-cell late-only patching is therefore not always a stable summary of the PT→IT change; it must be checked against the upstream context in which the late stack normally operates.
2. A reusable paired-checkpoint estimand. First-divergence factorial diffing tests portability directly at the natural PT/IT token disagreement by estimating the upstream $\times$ late interaction on the divergent-token margin. Across six dense PT/IT pairs, the interaction is positive in every family, remains positive without Gemma, survives the generated-position ≥ 3 check, and is label-aligned under a permutation null on the five-family raw-record subset. The OLMo-2 stage lineage shows where the estimand appears along one released post-training trajectory: the largest adjacent-stage jump is Base→SFT, followed by smaller positive SFT→DPO and DPO→RLVR interactions.

3. A graded depth-route characterization of the change. Candidate identity is more mid-sensitive: middle-positioned MLP substitutions transfer first-divergent token identity more often than late-positioned substitutions. Final margin/readout is more late-sensitive: late-positioned substitutions have larger margin effects, and late IT MLPs provide the strongest local IT-token support.

This does not claim to discover that late computation reads upstream state. The narrower contribution is to turn that working assumption into a falsifiable paired-checkpoint diagnostic with explicit nulls and stress tests. To our knowledge, prior paired-checkpoint diffing work (Du et al. 2025; Prakash et al. 2024; Lindsey et al. 2024; Minder et al. 2025) does not measure the upstream $\times$ late interaction at the natural first-token disagreement. The protocol applies to any paired-checkpoint contrast where the two checkpoints share architecture and tokenizer, including base $\rightarrow$ SFT, SFT $\rightarrow$ DPO, safety-tuned, reasoning-tuned, and constitution-modified model pairs.

We use IT throughout as readable shorthand for instruction-following post-trained descendants of released pretrained checkpoints; the recipes are heterogeneous (SFT, DPO, RLVR, multi-stage post-training), so the claim is about paired PT-versus-post-trained model diffs rather than a single training algorithm.

2. Setup

Terminology used throughout the paper.

Term	Meaning		
PT / IT	Pretrained checkpoint and its instruction-following post-trained descendant from the same model family.		
PT/IT pair	One paired pretrained/post-trained checkpoint contrast.		
Dense-6 core set	The six dense PT/IT pairs used for the core first-divergence factorial synthesis: Gemma 3 4B, Llama 3.1 8B, Qwen 3 4B, Mistral 7B, OLMo 2 7B, and Qwen2.5 32B.		
Dense-5 support set	The five 4B-8B dense families used in supporting analyses that were not rerun or not pooled with Qwen2.5-32B: Gemma 3 4B, Llama 3.1 8B, Qwen 3 4B, Mistral 7B, and OLMo 2 7B.		
First-divergence prefix	Earliest shared-history position where PT and IT prefer different next tokens (t_{PT} vs t_{IT}).		
Late stack	Transformer blocks from the pre-specified late boundary to the final hidden state.		
Late-MLP window	MLP sublayers inside the pre-specified late window; used in graft/swap, identity/margin, and write-out tests.		
Early / middle / late windows	Overlapping depth windows at comparable normalized depths across families (Appendix F gives boundaries). The middle/late identity-margin split in §3.3 is a relative-localization signal ...		
Divergent-token margin	$Y_R(U, L) = \text{logit}(t_{IT}) - \text{logit}(t_{PT})$, scored under a fixed readout R.		
Upstream $\times$ late interaction	$[Y(U_{IT}, L_{IT}) - Y(U_{IT}, L_{PT})] - [Y(U_{PT}, L_{IT}) - Y(U_{PT}, L_{PT})]$, the primary estimand.		
Final-20% KL region	Final normalized-depth segment used to summarize 'KL(layer		own final)' in §3.2.

2.1 Models

The core first-divergence evidence covers six dense PT/IT pairs. The five 4B-8B members are:

Model	Layers	d_model	Attention	Pre-training data	Post-training
Gemma 3 4B	34	2560	GQA, hybrid local/global (5:1)	Undisclosed	Multi-stage post-training

Model	Layers	d_model	Attention	Pre-training data	Post-training
Llama 3.1 8B	32	4096	GQA, all global	15T tokens	Iterative supervised + preference optimization
Qwen 3 4B	36	2560	GQA, all global	36T tokens, 119 languages	Multi-stage post-training
Mistral 7B v0.3	32	4096	GQA, full attention (sliding_window=null)	Undisclosed	Instruct checkpoint
OLMo 2 7B	32	4096	MHA, all global	OLMo-mix-1124	SFT + DPO + RLVR

The sixth dense member is larger:

Model	Layers	d_model	Attention	Role
Qwen2.5 32B	64	5120	GQA, all global	Included in the Dense-6 core first-divergence synthesis.

The exact Hugging Face checkpoint IDs used by the Dense-6 core runs are:

Family	PT checkpoint	IT checkpoint
Gemma 3 4B	google/gemma-3-4b-pt	google/gemma-3-4b-it
Llama 3.1 8B	meta-llama/Llama-3.1-8B	meta-llama/Llama-3.1-8B-Instruct
Qwen 3 4B	Qwen/Qwen3-4B-Base	Qwen/Qwen3-4B
Mistral 7B v0.3	mistralai/Mistral-7B-v0.3	mistralai/Mistral-7B-Instruct-v0.3
OLMo 2 7B	allenai/OLMo-2-1124-7B	allenai/OLMo-2-1124-7B-Instruct
Qwen2.5 32B	Qwen/Qwen2.5-32B	Qwen/Qwen2.5-32B-Instruct

These checkpoint IDs and their pinned Hugging Face snapshot revisions are part of the estimand: the claim is about these released PT/post-trained descendants, not an abstract training algorithm. The native layerwise curves use each model in its native prompting regime; raw-shared first-divergence and residual-state runs force identical raw prompt token IDs before comparing residual states. Appendix G records the immutable revisions now used by the experiment loaders, tokenizer notes, chat-template behavior, and the historical limitation that earlier traces stored repo IDs rather than resolved snapshot hashes.

Grouped rows labeled Dense-6 include Qwen2.5-32B. Rows labeled Dense-5 or Dense-4 deliberately exclude it because the corresponding supporting analysis was only run or only pooled for the five 4B-8B families. The Qwen2.5-32B PT/IT run repeats the core first-divergence factorial, identity/margin, MLP write-out, and raw-KL bridge analyses; its standalone details are reported in Appendix A.7, and its interaction is included in the Dense-6 §3.1 estimate. A non-pooled OLMo-2 7B stage-progression case study applies the same first-divergence factorial to released Base, SFT, DPO, and Instruct/RLVR checkpoints; it is reported in §3.1 and Appendix A.8 as local transition evidence, not as an additive decomposition of the full Base→Instruct/RLVR contrast.

2.2 Readouts and interventions

The implementation is model-adapter based rather than feature-dictionary based. Matched-prefix graft/swap and first-divergence tests operate on raw MLP activations and residual-stream states through a shared adapter interface, and no core claim depends on transcoders, SAEs, or model-specific decomposition dictionaries. The reported residual interventions are restricted to paired PT/IT checkpoints with compatible residual dimensions, tokenizer IDs for the tested prompts/tokens, and block semantics; we do not patch residual states across unrelated architectures.

Definition 1 fixes the core notation. The setup section adds the implementation details needed to interpret the interventions:

Object	Implementation detail	Role in the story		
First-divergence factorial diffing	Patch upstream residual state at the late boundary, run PT or IT downstream blocks at the first-divergence ...	Estimates the primary upstream-late interaction.		
Identity/margin decomposition	Measure token transfer separately from the IT-vs-PT logit margin at the same shared prefix.	Separates candidate identity from final readout pressure.		
Delayed-stabilization context	Track $\text{KL}(p_L)$		p_L)` from each layer's decoded distribution to the same model's final distribution.	Motivates testing late windows under controlled histories.

Prompting is matched to the question being asked:

Analysis	Prompt regime	Why this regime is used
First-divergence factorial and identity/margin tests	Raw-shared: PT and IT receive identical raw prompt token IDs before the divergent-token comparison.	Isolates checkpoint computation under a shared history.
Native layerwise convergence curves	Native: PT uses raw prompt text; IT uses the checkpoint chat template.	Measures the deployment-style layerwise signature that motivated late-window interventions.
Matched-prefix graft/swap and JS replay	Teacher-forced shared histories under the run's specified raw or native prompt stream.	Tests late-window effects after removing generated-history divergence.

First-divergence position distribution. First-divergence prefixes are selected events, not random token positions. In the five-family 4B-8B holdout, the 2,983 valid events occur early: mean generated position 2.17, median 0, with 1,499 at the first generated token. Qwen2.5-32B contributes 1,397 additional valid events with a later median position (2), giving 4,380 Dense-6 prompt clusters for the core synthesis. Position 0 is still the next-token prediction after the full raw prompt under identical raw token IDs, not a BOS-only or chat-template comparison. The five-family IT-side divergent tokens are mixed (1,265 content, 685 format, 1,033 function/other), and 551/2,983 events contain an assistant-marker token. We treat position as an informative axis of the phenomenon: first-response disagreements, later-content disagreements, and format/safety/register disagreements are different regimes of post-training behavior. Section 3.1 therefore reports position-stratified and category-mix checks; Appendices A.7 and F give the full distribution notes.

The late boundary in the Dense-6 first-divergence factorial is the first layer of the pre-specified late window: Gemma layer 20, Llama layer 19, Qwen3 layer 22, Mistral layer 19, OLMo layer 19, and Qwen2.5-32B layer 38 (late window 38-63). Appendix F lists the five 4B-8B windows and boundary-sensitivity checks; Appendix A.7 reports the Qwen2.5-32B details.

Implementation details matter for interpretation but are mechanical: residual-state patching replaces the full hidden-state sequence at the late boundary and recomputes downstream blocks; the diagonal 2×2 cells are asserted to reconstruct native PT/IT passes; and the off-diagonal cells are the only non-trivial hybrids. Appendix G gives the exact tolerance, clustering rule, and audit paths.

Supporting machinery: the delayed-stabilization signature $KL(p_L || p_L)$ for layerwise context, same-history replay and endpoint matching (coarsened exact matching, lacus et al. 2012, on final entropy and top-1/top-2 margin) for confound control, label-swap permutation nulls, and matched random residual-projection controls for the late-MLP KL analysis (per Heimersheim and Nanda 2024). Each is introduced only where it protects a claim.

All causal language is intervention-scoped: replacing an upstream state, late stack, or MLP window in a constructed forward pass changes a specified readout. The protocol does not assert that hybrid states are natural states or that any intervention recovers a feature-level circuit; matched histories, label-swap controls, raw-shared validation, and the matched random control reduce obvious artifacts but do not remove the need to interpret each intervention with its metric and direction fixed.

Secondary diagnostics are demoted to appendices. Commitment summaries and raw-lens variants address metric/probe robustness; δ -cosine and feature-level Gemma analyses are supplementary diagnostics, not part of the evidence spine.

2.3 Evidence sets

The core Dense-6 first-divergence results use controlled-history runs that freeze the token history, compare pure and intervened PT/IT branches, and supply the upstream-late factorial. The five 4B-8B runs use 400- and 600-prompt subsets from the same broad prompt pool; Qwen2.5-32B uses its own larger prompt set and is pooled into the Dense-6 factorial synthesis. Supporting identity/margin decomposition, same-history JS, graft/swap, and MLP write-out analyses are labeled by their actual scope: Dense-6 when Qwen2.5-32B is included, Dense-5 when only the five 4B-8B families are pooled.

Layerwise context comes from two supporting runs: a native free-running 2,936-prompt convergence analysis, and a dense-family 600-prompt endpoint-matched run with raw and tuned probes.

2.4 Code and artifact availability

For double-blind review, code and paper-facing artifacts are released through an anonymized archive containing the model adapters, experiment packages, launch scripts, analysis scripts, prompt datasets, summary tables, bootstrap intervals, human-audit summaries, and final plots needed to audit the claims.

The release commits the summaries and plots from which manuscript numbers are read, plus a mechanical audit endpoint, `bash scripts/reproduce/reproduce_claims_from_summaries.sh`, that checks the primary numbers against those artifacts. Large raw activation arrays, probe tensors, tuned-lens checkpoints, and per-token traces are omitted from git for size

and mirrored separately where needed. Appendix G maps each main claim to commands, expected artifacts, expected numbers, and rerun costs.

Internal run identifiers and file-level provenance are kept in Appendix G and the anonymized artifact archive rather than in the main narrative.

3. Results

The results start with the primary first-divergence factorial. The question is not whether late layers read upstream state in the ordinary architectural sense; they do. The question is whether a late-only PT→IT patch is a stable model-diffing summary, or whether its effect changes with the upstream residual state. We then give the delayed-stabilization context that made late windows worth testing, followed by the first-divergence identity/margin decomposition.

3.1 Core diagnostic: late-stack effects are not portable across upstream states

**TL;DR.* The same late stack has a much larger effect on the divergent-token margin from an IT-shaped upstream state than from a PT-shaped one. The asymmetry is positive in every Dense-6 core family, remains positive without Gemma, and remains positive at generated position ≥ 3 with all six family intervals above zero. It is label-aligned in the five-family raw-record null ($p=5 \times 10^{-5}$). On factual/reasoning prompts, the simple late-only term reverses sign while the factorial interaction stays positive, so the one-cell late-only patch is not a stable summary of the PT→IT change.*

Using Definition 1 with a fixed readout R , the simple late effect from PT upstream is $Y_R(U_{PT}, L_{IT}) - Y_R(U_{PT}, L_{PT})$; the simple late effect from IT upstream is $Y_R(U_{IT}, L_{IT}) - Y_R(U_{IT}, L_{PT})$; their difference is the upstream $\times$ late interaction, the paper's primary estimand. This section reports both the factorial interaction and the one-cell late-only term because their disagreement is the diagnostic: if late-only patching were a portable summary, the two would tell the same story. The common-IT readout scores every cell with the IT final norm, lm_head , and real-token mask; the common-PT variant uses the PT readout analogously; native readouts score each cell with its host checkpoint's own readout.

The 2×2 decomposition. Under the common-IT readout, swapping in the IT late stack shifts the IT-vs-PT margin by $+0.639$ logits (95% CI $[+0.570, +0.709]$) from a PT upstream state, but by $+3.076$ logits (95% CI $[+2.978, +3.174]$) from an IT upstream state. For the Dense-6 core set, the matched 2×2 decomposition gives a late-stack main effect of $+1.858$ $[+1.786, +1.930]$, a larger upstream-context effect of $+3.874$ $[+3.761, +3.987]$, and a positive interaction of $+2.437$ $[+2.353, +2.521]$. Thus the IT late stack is not a portable late delta: the same downstream computation contributes much more IT-token margin when the upstream state is already IT-shaped. This is the paper's central model-diffing estimate.

Common-PT readout cross-check. Scoring every cell with the PT readout instead gives the same direction and magnitudes ($+0.662$, 95% CI $[+0.600, +0.724]$, versus $+3.083$, 95% CI $[+2.986, +3.180]$, for IT-late given PT versus IT upstream; interaction $+2.421$, 95% CI $[+2.337, +2.506]$), confirming the result is not a readout-choice artifact.

Dissociation from the simple late-only effect. On a factual/reasoning stress test, the IT late stack from PT upstream moves **against** the IT token (-1.18 logits, 95% CI $[-1.26, -1.09]$), while the upstream $\times$ late interaction remains positive ($+1.81$ logits, 95% CI $[+1.72, +1.90]$). This is the failure mode the factorial is meant to catch: the simple late-only term tracks prompt regime, while the four-cell interaction continues to measure whether late computation is portable across upstream states.

Appendix F gives the factual/reasoning, per-family, and subgroup breakdowns.

The practical diagnostic is the gap between two late-stack effects: the one-cell late-only patch from PT upstream, and the matched-context late effect from IT upstream. We call this the late-effect portability gap in the table below; algebraically it is the same upstream $\times$ late interaction, not a new estimand. The scope column matters: the Dense-6 row is the core family-balanced synthesis, while subgroup rows use the five-family artifacts that carry prompt, token, and position labels. The headline Dense-6 generated-position ≥ 3 interaction is $+1.43$; the $+1.52$ row below is the corresponding Dense-5 labeled-position audit.

Regime	Scope	Late-only from PT upstream	Matched-context late effect	Portability gap / interaction	Diagnostic read
First-divergence	Dense-6 family-balanced core synthesis	$+0.64$	$+3.08$	$+2.44$	Late-only underestimates matched-context late computation.
Factual/reasoning extension	Dense-5 content/reasoning extension	-1.18	$+0.64$	$+1.81$	Sign failure for the late-only summary.
Later positions ≥ 3	Dense-5 labeled-position audit	$+0.98$	$+2.49$	$+1.52$	Underestimates in later shared-history regimes.

Regime	Scope	Late-only from PT upstream	Matched-context late effect	Portability gap / interaction	Diagnostic read
Governance/conversation	Dense-5 subgroup-labeled records	+0.64	+2.69	+2.05	Underestimates.
Format/register	Dense-5 subgroup-labeled records	+0.76	+4.37	+3.61	Strong underestimation.
Safety	Dense-5 subgroup-labeled records	+0.25	+3.08	+2.83	Strong underestimation.

Label alignment. A label-swap control asks whether the sign is PT/IT-oriented rather than an arbitrary orientation. On the five-family raw-record subset, preserving each prompt's four factorial cell values but randomly swapping PT/IT labels gives a null centered near zero (99.9th percentile +0.239 logits), while the observed Dense-5 interaction is far outside it (+2.64, $p=5.0e-5$ with 20,000 permutations). This does not replace the Dense-6 interval evidence above; it rules out a metric-construction artifact in the raw-record subset where the null can be recomputed directly.

Position regimes. In the Dense-6 core set, the interaction remains positive after removing immediate first-token divergences (+2.08, 95% CI [+1.96, +2.19]) and through generated position ≥ 3 (+1.43, 95% CI [+1.30, +1.57]; all six family-specific intervals above zero). Position selects different disagreement regimes — format and safety openings, mid-response content disagreements, later shared-history disagreements — and the magnitude differences across them are part of the result. Position 0 is not a chat-template or BOS artifact: the factorial uses raw-shared prompts on both branches, validates identical token IDs, and evaluates the first generated token after the full prompt-conditioned residual state. Per-position thresholds appear in the headline-robustness table below; the five-family category-mix audit for the ≥ 3 subset is in Appendix F, and the Qwen2.5-32B audit is in Appendix A.7.

Figure 2: Core diagnostic: late-stack effects are not portable across upstream states under this readout. First-divergence factorial diffing crosses upstream residual state with downstream late stack while holding the scoring readout fixed. The left panel shows the Dense-6 common-IT 2x2 contrast: the IT late-stack effect is +0.64 logits from PT upstream and +3.08 logits from IT upstream, producing an upstream-state x late-stack interaction of +2.44 logits (95% CI [+2.35, +2.52]). The right panel shows family-level interaction CIs for all six dense pairs; each per-family 95% interval excludes zero, and the Gemma-removed Dense-5 estimate remains positive (+1.71 logits, [+1.64, +1.78]). A conservative generated-position ≥ 3 check remains positive (+1.43 logits, [+1.30, +1.57]) with all six family intervals above zero, showing the result is not only an immediate response-opening effect. The label-swap null is computed on the five-family raw-record subset, where the observed Dense-5 interaction is far outside the null 99.9th percentile (+0.239 logits; $p=5.0e-5$). The factual/reasoning stress test reported in text is the companion validity check: the one-cell late-only term flips sign while this interaction remains positive. We do not use the IT/PT ratio as the inferential quantity because the small PT-upstream denominator inflates ratios. Magnitude varies by family and by first-divergence position (Section 3.1, Appendix F).

Families are sorted ascending by interaction magnitude; the IT/PT ratio is omitted because the small PT-upstream denominator inflates ratios (descriptive only) and the logit interaction is the inferential quantity.

Family	First-divergence records	IT late stack from PT upstream	IT late stack from IT upstream	Interaction (95% CI)
Llama 3.1 8B	600	+0.79	+2.05	+1.25 [+1.10, +1.42]
Qwen2.5 32B	1,397	+0.98	+2.42	+1.45 [+1.32, +1.57]
Qwen 3 4B	600	+0.59	+2.05	+1.46 [+1.32, +1.62]
OLMo 2 7B	586	+0.76	+2.61	+1.85 [+1.67, +2.03]
Mistral 7B	597	+0.61	+3.15	+2.53 [+2.35, +2.72]
Gemma 3 4B	600	+0.10	+6.18	+6.08 [+5.72, +6.44]
Dense-6 (family-balanced)	4,380	+0.64	+3.08	+2.44 [+2.35, +2.52]
Dense-5 (Gemma removed)	3,780	+0.75	+2.46	+1.71 [+1.64, +1.78]

Family heterogeneity, including Gemma. The interaction is positive in all six dense pairs, but the magnitude is not uniform: Llama is smallest (+1.25) and Gemma is largest (+6.08). The +2.44 headline is therefore a family-balanced Dense-6 mean, not a claim that families share one scalar effect size. Conservative summaries remain positive: Gemma-removed Dense-5 +1.71, family median +1.66, and trimmed mean +1.82. Appendix A.1 centralizes the Gemma diagnostics; we report Gemma as a high-magnitude family rather than explaining it away with a single weight-change story.

Subgroups. In the five-family holdout with detailed category labels, the interaction is positive for every prompt-category subgroup: conversational/governance (+2.05), format (+3.61), safety (+2.83), and for every first-divergent IT-token type: content-like

(+2.50), format (+2.60), function/other (+2.81). These strata characterize where the estimate comes from rather than adding new claims; Qwen2.5-32B category notes are in Appendix A.7.

Headline robustness (the interaction). In the Dense-6 core set, the interaction is positive across every family aggregation and position threshold we test. Magnitude variation across these axes is part of the characterization: early response formation and Gemma have larger effects, while later-position and Gemma-removed estimates remain positive. Rows labeled Dense-5 are intentionally five-family diagnostics where the underlying artifact does not include Qwen2.5-32B.

Check	Result	Interpretation
Dense-6 primary (family-balanced)	+2.44 [+2.35, +2.52]	Family-balanced over all six dense pairs.
Gemma removed (Dense-5)	+1.71 [+1.64, +1.78]	Conservative summary; not only a Gemma effect.
Family median	+1.66	Typical-family magnitude across six dense pairs.
Trimmed family mean	+1.82	Drops smallest and largest family interactions.
Drop generated position 0	+2.08 [+1.96, +2.19]	Not only an immediate-disagreement effect.
Generated position ≥ 3	+1.43 [+1.30, +1.57]	All six family-specific intervals above zero.
Generated position ≥ 3 , Gemma removed	+0.83 [+0.75, +0.91]	Later-position regime remains positive without Gemma.
Generated position ≥ 5	+1.48 [+1.28, +1.68]	Lower-powered later-position regime.
Generated position ≥ 5 , Gemma removed	+0.80 [+0.70, +0.90]	Llama family interval compatible with zero in the five-family subset.
Factual/reasoning stress test (Dense-5)	+1.81 [+1.72, +1.90]	Persists outside governance/register-heavy holdout.
OLMo-2 staged lineage	Base→SFT +0.78; SFT→DPO +0.14; DPO→RLVR +0.016; Base→RLVR +1.93	Non-pooled local-transition case study over released OLMo stages.
Label-swap null (Dense-5 raw records)	observed +2.64; null 99.9% +0.239; $p=5 \times 10^{-5}$	Interaction is PT/IT-label aligned where the raw null is recomputed.

All rows above use the common-IT 2×2 interaction readout. The diagnostic table separates the PT-upstream late-only term from the matched-context late effect; the robustness table below focuses on the interaction itself.

The OLMo-2 staged-lineage case study checks a different scope question: how the same estimand changes across released post-training stages. The interaction is already substantial at Base→SFT, remains positive but smaller for SFT→DPO and DPO→RLVR, and is largest for the full Base→RLVR contrast. This supports the interpretation that post-training creates or amplifies a middle-to-late compatibility relation rather than merely exposing a generic transformer dependency. Appendices A.7-A.8 give the Qwen2.5-32B and OLMo details.

Companion controls (identity/margin decomposition and late-window context).

Check	Result	Interpretation
Raw-shared PT-host IT-token transfer (Dense-5 support set)	middle 26.0% [24.5%, 27.7%]; late 17.6% [16.2%, 18.9%]	Identity split survives without native IT templates; Qwen2.5-32B repeats the same ordering in Appendix A.7.
Native IT-template IT-host margin drop (Dense-5 support set)	late 13.25 logits [12.91, 13.61]; middle 12.01 [11.66, 12.35]	Native prompting makes late margin/readout strongest.
Matched random residual projection (Dense-5 support set, final-20% KL)	true late graft +0.327 nats [+0.298, +0.359]; random +0.003 [-0.002, +0.008]	Late MLP graft is not a generic late-window perturbation.

3.2 Layerwise context: delayed stabilization and late-window localization

TL;DR. Context for why we tested late windows: IT models stay farther from their own final-layer prediction than PT models do. Late-MLP grafts have the largest tested leverage on that delay, with specificity confirmed against a matched random control (+0.327 nats vs +0.003). This section motivates the late boundary used by the factorial; it is not the causal headline.

The §3.1 factorial result is computed on raw logits at the divergent prefix and does not depend on a learned probe; this section provides the layerwise context for why we tested late windows in the first place. Under native free-running decoding, IT models remain farther from their own final next-token distribution than PT models do through much of the forward pass. Under the tuned lens, the dense-family IT-minus-PT KL (layer || own final) gap is positive in the early, middle, and late thirds of the network (+0.62, +0.54, and +0.33 nats), and raw-lens variants preserve the qualitative ordering.

Intervals in this section are family-bootstrap (over dense-family means) for KL graft/swap depth ablations and prompt-bootstrap for random-control rows; they support the layerwise context and are not the same bootstrap object as the §3.1 factorial intervals.

Layerwise Delayed Stabilization in Five Dense PT/IT Pairs

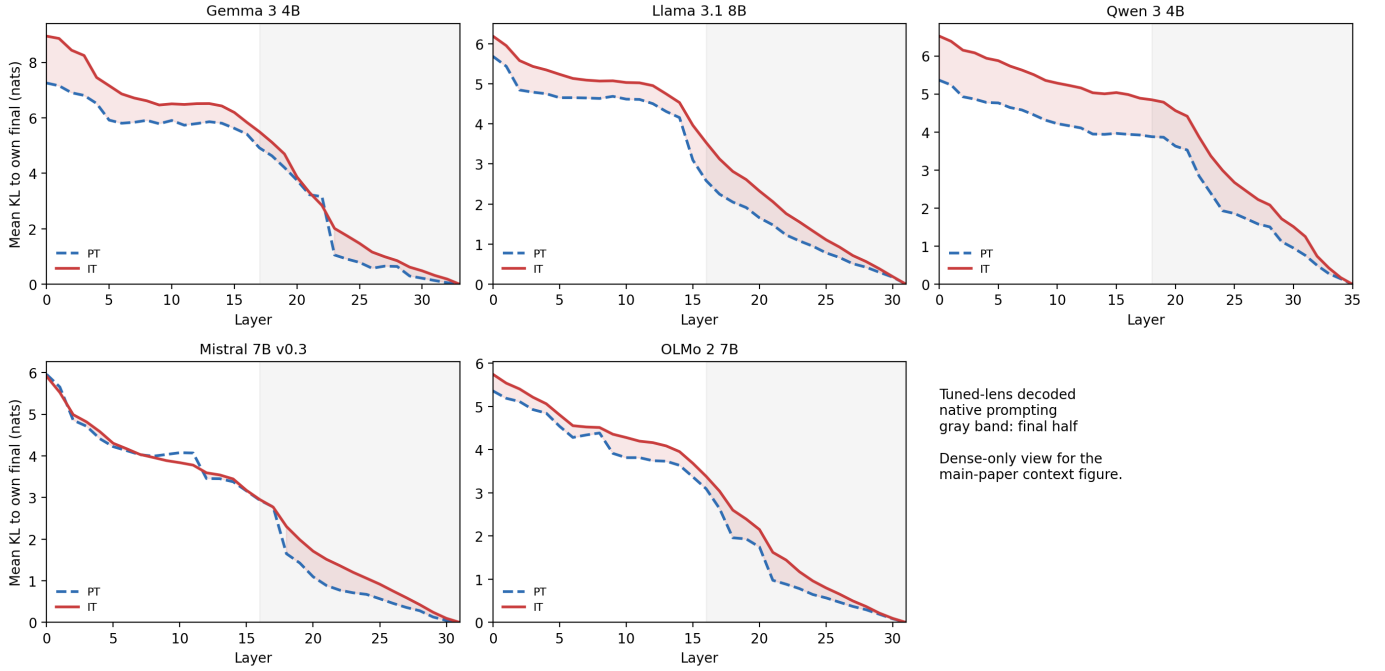


Figure 3: Layerwise context: delayed stabilization. Across the five dense families, IT models stay farther from their own final-layer prediction than PT models do across much of the forward pass. Mean $KL(\text{layer} \parallel \text{final-layer distribution})$ per layer, tuned-lens decoded. This layerwise pattern led us to test late-window effects under matched-prefix interventions and first-divergence factorials.

This metric is endpoint-relative, so we check it against the most direct endpoint and history confounds. After matching token steps on final entropy, final confidence, and final top-1/top-2 margin, the late IT-minus-PT gap remains positive under both raw (+0.425 nats, 95% CI [+0.356, +0.493]) and tuned (+0.762 nats, 95% CI [+0.709, +0.814]) probes. Under identical teacher-forced histories, same-layer PT/IT JS divergence is also positive and grows late (0.121 pre-late to 0.196 final-20% under the prompt-mean regional estimator). Appendix A gives the full endpoint-matching, same-history replay, probe, commitment, per-family, and reverse-teacher views.

Under identical token histories, late MLP substitutions have the largest tested leverage on that delayed-stabilization side. On the dense-family mean, the final-20% KL effect is +0.34 nats (family-bootstrap 95% CI [+0.18, +0.50]) for PT with late IT MLPs grafted in, versus -0.03 (95% CI [-0.10, +0.02]) early and -0.05 (95% CI [-0.11, +0.02]) middle. The raw late-graft effect is positive in all five dense families (+0.115 to +0.609 nats), and leave-one-family-out dense means remain positive (+0.274 to +0.398), so the result is not driven by a single family.

The mirrored swap test asks whether removing late IT MLPs from an IT host also reduces the delay. It does: replacing late IT MLPs with PT MLPs produces the largest reduction of IT delayed stabilization, with a dense-family mean effect of -0.51 nats (family-bootstrap 95% CI [-0.83, -0.22]) versus -0.10 (95% CI [-0.26, +0.03]) early and -0.23 (95% CI [-0.37, -0.09]) middle.

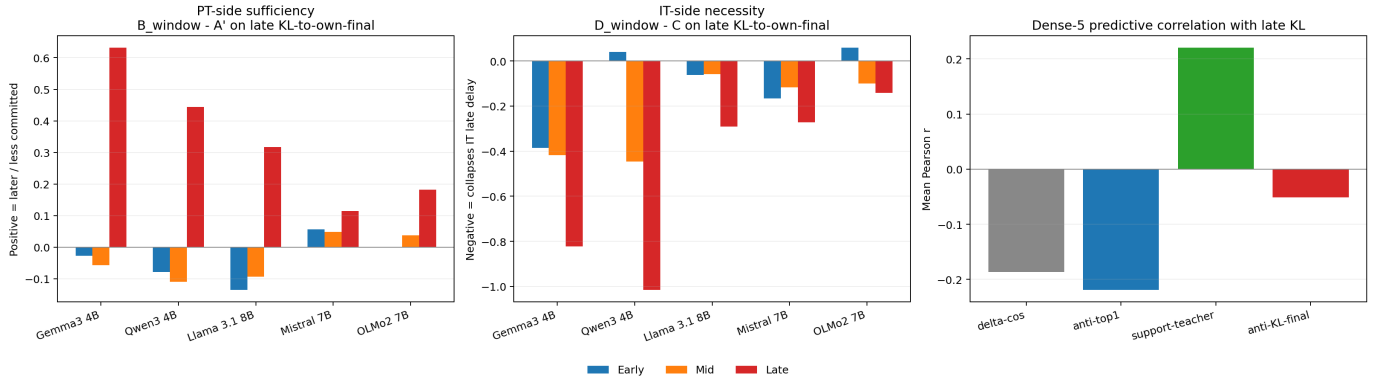


Figure 4: Symmetric graft/swap localization. Left and center: late MLP substitutions have the largest convergence-gap effect in both directions, increasing the delay in a PT host and reducing it in an IT host. The matched random-control number in the text checks that this is not a generic late-window perturbation effect.

A matched random-control follow-up rules out the simplest "late layers are fragile" reading. Replacing the learned late IT-minus-PT MLP effect with matched random residual-projection controls gives a dense-family final-20% KL effect of $+0.003$ nats ($[-0.002, +0.008]$), compared with $+0.327$ ($[+0.298, +0.359]$) for the true late graft — a true-minus-random margin of $+0.324$ nats ($[+0.294, +0.358]$). Late MLP substitutions are therefore the only tested substitutions with a large delayed-stabilization effect, not generic perturbations of a sensitive window. Appendix A gives per-family, probe, weight-change, and random-control details.

3.3 Supporting decomposition: middle-positioned substitutions transfer identity more often than late substitutions

TL;DR. Within the Dense-5 support-set decomposition, middle-positioned MLP substitutions transfer token identity more often than late-positioned (26% vs 18%; both well below 50%, so neither window alone controls token identity — this is a relative-localization signal). Late-positioned substitutions show larger margin/readout effects. Qwen2.5-32B repeats the same middle-over-late identity ordering in Appendix A.7, but the full table below keeps the common Dense-5 bootstrap scope.

The factorial estimates the upstream $\times$ late interaction; we now ask what the middle and late windows contribute to first-divergence token formation. At prefixes where PT and IT first choose different next tokens, which depth window changes the token identity and which changes the final margin? The first-divergence test evaluates intervened models at the shared prefix, and the companion MLP write-out test measures each window's finite-difference logit support for the PT token, the IT token, and top alternatives.

Identity and margin readouts are reported in one table so the two effects are not conflated; intervals are 95% percentile bootstrap over the Dense-5 support-set prompt clusters or prompt-level first-divergence records. The middle and late windows overlap by 3–4 layers per family (Appendix F), so the comparison below is between *overlapping* depth windows at comparable normalized depths, not between disjoint subnetworks; the asymmetry should be read as a relative-localization signal rather than as evidence for non-overlapping module assignments.

Readout	Early MLP window	Middle MLP window	Late MLP window	What it says
PT host: IT-token transfer	—	26.0% [24.5%, 27.7%]	17.6% [16.2%, 18.9%]	Middle substitutions transfer candidate identity more often.
IT host: PT-token transfer	—	31.2% [29.6%, 32.9%]	20.8% [19.4%, 22.3%]	Mirror direction gives the same identity pattern.
IT-host margin drop	11.53 [11.20, 11.88]	12.01 [11.66, 12.35]	13.25 [12.91, 13.61]	Late substitutions have the largest margin effect.
Pure IT MLP support for t_{IT}	-0.041 [-0.049, -0.032]	+0.021 [+0.011, +0.032]	+0.789 [+0.754, +0.825]	Native IT-token support is concentrated late.
PT-to-IT change in t_{IT} support	+0.034 [+0.027, +0.042]	+0.070 [+0.059, +0.080]	+0.715 [+0.683, +0.747]	The learned write-out difference is also concentrated late.

The first two rows give the identity side. The transfer rates are well below 50%, so a single-window MLP substitution does not dominantly redirect token identity. The middle-over-late gap is a relative localization signal: middle windows are systematically

more involved in candidate identity than late windows, not a claim that any one window alone controls the divergent token.

The last three rows show why the late window still matters. Late MLP windows have the largest effect on the IT-vs-PT margin and provide the strongest local support for the IT divergent token in native IT runs. The local MLP-only proxy is included as a miniature version of the §3.1 portability check: inside the native IT trajectory the late MLP update strongly supports t_{IT} , but transplanting late IT MLP updates into a PT host changes the same fixed-prefix IT-vs-PT margin by only $+0.0035$ logits (95% CI $[-0.001, +0.009]$). Thus the write-out rows should be read as native late support, not as evidence for a portable MLP-only module.

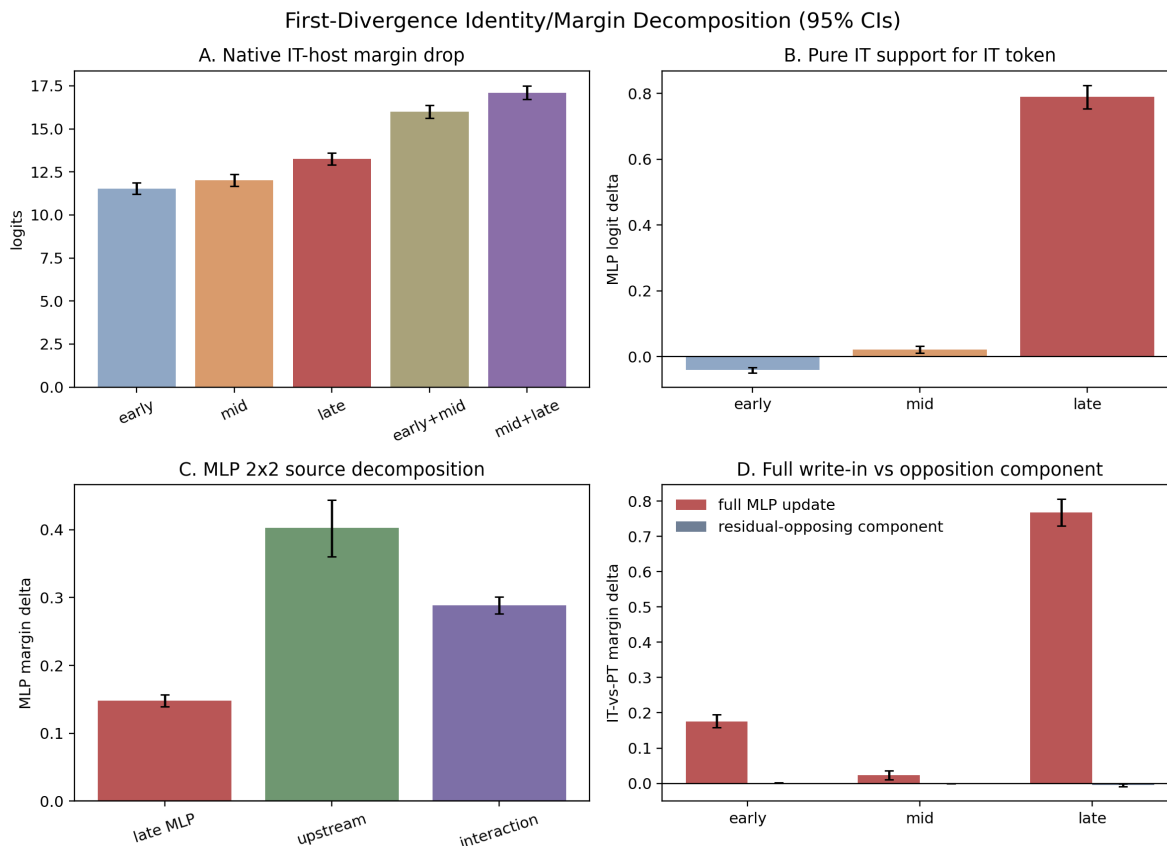


Figure 5: Candidate transfer and local late support. First-divergence tests show that middle substitutions transfer PT-vs-IT token identity more, while late IT MLP windows provide the strongest native IT-token support. The full late MLP update carries the IT-vs-PT margin write-in in this proxy.

Putting both sides together: middle windows are more diagnostic of which PT- or IT-like token candidate is exposed, while late windows interact with upstream state to shape the final margin. Mid and late windows must be modeled jointly under this readout — neither in isolation captures the PT/IT disagreement. The delayed-stabilization pattern in §3.2 is the layerwise trace of this handoff: upstream state already encodes much of the PT/IT decision, and late computation changes the margin only in compatibility with that state. Appendix A gives the token-flow chronology and additional same-history controls.

4. Related Work: From Stage Heuristics to Estimands

What prior work does not provide. Prior work makes late-stage dependence plausible, but it does not give a diagnostic for whether a late-only patch is portable across upstream states at the actual PT/IT disagreement token. Our contribution is the estimand: a paired PT/IT upstream-state $\times$ late-stack interaction at the first natural token where the checkpoints disagree, with a label-swap permutation null for the factorial and a matched random residual-projection control for the late-MLP localization. Du et al. (2025) is the closest cross-family comparator: they compare base and post-trained checkpoints globally on knowledge, truthfulness, refusal, and confidence; we condition on the natural first-token disagreement and run hybrid-pass interventions at that exact prefix. Prakash et al. (2024) is the closest methodological precedent (cross-model activation patching across base and fine-tuned models); their target is entity tracking and ours is the natural PT/IT next-token disagreement under matched-prefix control. Following the caution urged by activation-patching work (Heimersheim and Nanda 2024), we describe these interventions as effects on measured readouts, not as complete mechanism recovery.

What the estimand rules out. Since transformer layers are trained end-to-end, ordinary upstream dependence is not the claim and would not be novel. The model-diffing claim is narrower: the PT→IT change does not behave like a portable late update whose IT-token margin contribution is similar across PT-shaped and IT-shaped upstream states. The factual/reasoning stress test makes this sharper: the simple late-only term can move against the IT token while the upstream × late interaction remains positive. This is not a general critique of patching; it is a warning that a one-cell late-only patch can be the wrong summary statistic for a paired-checkpoint contrast. The matched random control also matters in this narrower sense: a matched random late-MLP perturbation does not reproduce the delayed-stabilization effect (+0.003 vs +0.327 nats). What remains for circuit-level work is feature mediation: which upstream features make the IT late stack effective, and which late subspaces carry the remaining interaction?

Compatibility with the layerwise and intervention literatures. The vocabulary for late-stage dependence — residual-stream computation (Elhage et al. 2021), FFN readouts (Geva et al. 2022a,b), tuned-lens prediction refinement (nostalgebraist 2020; Belrose et al. 2023), DoLA contrastive decoding (Chuang et al. 2024), late-stage residual sharpening and confidence calibration (Lad et al. 2025; Joshi et al. 2025), and task-oriented layer transitions (Zhao, Ziser, and Cohen 2024; Panigrahi et al. 2023) — presupposes upstream dependence. We operationalize the counterfactual at the natural first-divergent token rather than proposing a competing stage theory.

The middle/late identity-margin split echoes Zhao et al. and Panigrahi et al. with explicit counterfactual interventions, and is compatible with rather than competing with steering-vector and safety-localization work (Panickssery et al. 2024; Ardit et al. 2024; Li et al. 2025; Jain et al. 2024). Lin et al. (2024) document that PT/IT next-token distributions diverge most at stylistic, format, and safety-register positions — exactly where our first-divergence prefixes concentrate (Appendix F category mix). Our upstream × late interaction is the *mechanistic* counterpart of that surface observation: at the same positions where Lin et al. observe a distribution-shift signature, we measure whether late computation's effect on that shift is portable across upstream states, and find it is not. Sparse-crosscoder model diffing (Lindsey et al. 2024; Minder et al. 2025) gives a complementary feature-level route into paired-checkpoint comparisons — a natural follow-up is to train a crosscoder at the late boundary and test whether IT-specific features mediate the upstream × late interaction. First-divergent-token metrics also appear in model-compression work (Deiseroth et al. 2024) for detecting where a compressed model first departs from its reference; we instead condition on the first natural PT/IT disagreement to run paired interventions.

Taken together, the measurement battery supports a depth-window model-diffing account: post-training changes next-token formation through a middle-to-late handoff. The first-divergence factorial measures the handoff as a positive upstream × late interaction; the factual/reasoning stress test shows why the simple late-only term is not a stable substitute; the identity/margin decomposition makes middle windows more tied to candidate transfer than late windows; and delayed stabilization explains why late revision was the right place to test. The contribution is not another qualitative stage taxonomy, but a label-aligned, stratified counterfactual estimate at the natural first PT/IT next-token disagreement, with a reusable paired-checkpoint protocol.

5. Limitations and Next Tests

The result is a window-level model-diffing measurement. The interventions estimate effects on specified readouts in hybrid forward passes; they do not identify the features, heads, or complete set of MLP directions that mediate the upstream state. The next circuit-level test is feature- or subspace-level mediation inside the middle-to-late handoff, for example training a BatchTopK crosscoder at the late boundary and testing whether IT-specific features carry the interaction.

The empirical scope is six dense PT/IT pairs for the core first-divergence evidence: five 4B-8B families plus Qwen2.5-32B. Supporting layerwise, graft/swap, and random-control analyses remain Dense-5 where they were not rerun or pooled at 32B, and are labeled that way. We also include one non-pooled OLMo-2 7B stage-progression case study. This is broad for mechanistic model diffing, but it is not frontier-scale validation and it excludes non-dense claims from the main pool. The DeepSeek MoE run remains an appendix side case because dense MLP grafts and MoE routing perturbations are not the same intervention.

First-divergence events are selected token regimes, not random positions. That is the object of study: the protocol asks what happens at the first natural token where post-training changes the next-token preference. Position, prompt category, and token category stratification therefore characterize the effect rather than invalidate it. Native convergence-gap analyses are supporting context; the first-divergence factorial is the primary estimand.

6. Conclusion

Post-training does not appear in these paired checkpoints as a portable late-only delta. At the first token where PT and IT disagree, IT late computation contributes much more IT-token margin when the upstream residual state is already IT-shaped; the same pattern is positive across six dense PT/IT pairs. In the OLMO-2 stage lineage, the largest adjacent-stage jump appears at Base→SFT, with smaller positive SFT→DPO and DPO→RLVR interactions and a large Base→RLVR contrast. The factual/reasoning stress test shows the practical reason to use the factorial: a simple late-only patch can reverse sign even when the upstream $\times$ late interaction remains positive. Middle-positioned substitutions move candidate identity more often, and late-positioned substitutions move margin more. First-divergence factorial diffing turns that handoff into an auditable model-diffing diagnostic: a way to measure how a post-trained descendant reroutes next-token formation relative to its pretrained ancestor.

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Appendix Guide

The appendices are organized by purpose rather than by experiment ID. Appendix figures use local monotone labels (Figure A1-A40 in Appendix A, Figure B1 in Appendix B) rather than legacy run numbers; appendix tables use Table A1-A6. A reviewer checking a specific claim can use this map:

Need to check	Where to go	What is there
Primary upstream-late interaction and audit trail	Appendix G	Claim-to-command-to-number table for Exp23, label-swap null, subgroup checks, and content/reasoning extension.
Why late windows were tested	Appendix A.3-A.4	Probe/commitment robustness, endpoint matching, same-history JS, graft/swap localization, and matched random control.
Gemma magnitude caveat	Appendix A.1 and A.3	Weight-change diagnostics, raw-vs-tuned lens sensitivity, and Gemma-specific layerwise plots.
Identity/margin decomposition	Appendix A.5	Same-history JS, first-divergence token identity, margin flow, and candidate/amplification chronology.
Qwen2.5-32B dense-family member	Appendix A.7	Standalone details for the sixth Dense-6 core pair: Exp20/21/23, raw-KL bridge, and position audit.
OLMo-2 stage progression	Appendix A.8	Non-pooled Base/SFT/DPO/Instruct local-transition case study.
Auxiliary mechanism check	Appendix A.9	Natural-rollout controls that ablate the late residual-opposing MLP component while each model predicts its own generated tokens.
Auxiliary consequence check	Appendix B	LLM-judge design, human-audit resolved rates, unresolved-label rates, and kappa diagnostics.
Scope and boundary interpretation	Appendix F	Window definitions, first-divergence position sensitivity, prompt formatting, pooling, and mechanism granularity.
Reproducibility details	Appendix G	Checkpoint manifest, artifact policy, commands, hardware costs, and code/artifact map.

Auxiliary mechanism and consequence checks

Two appendices preserve extra experiments that are informative but not part of the paper's main evidence spine. Appendix A.9 asks whether the late residual-opposing MLP component matters during each model's own natural rollout. Appendix B asks whether the same intervention family has visible consequences under free-running natural decoding. Both are useful sanity and mechanism-adjacent checks; neither is used to establish the core first-divergence factorial claim.

Appendix A: Supplementary Evidence Map

This appendix keeps the full diagnostic surface, grouped by the role each artifact plays in the argument.

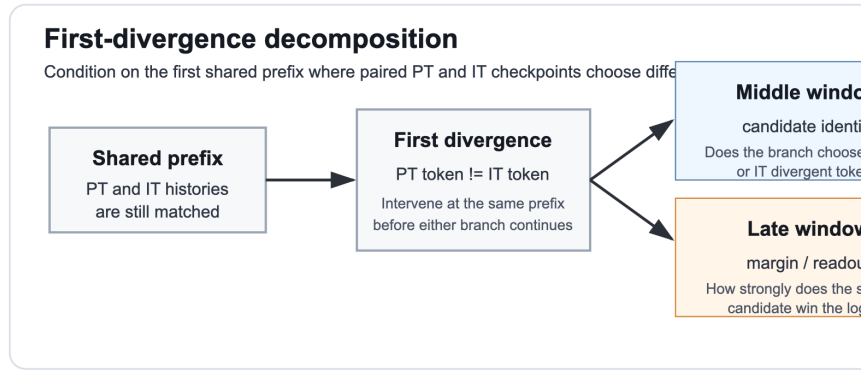


Figure A1: First-divergence factorial diffing schematic. A shared prefix reaches the first PT/IT next-token disagreement; middle-window interventions primarily test candidate identity, late-window interventions test margin/readout, and the first-divergence factorial estimates the upstream-late interaction.

A.1 Weight-change and auxiliary diagnostics

Figure A2: Generation-step $\times$ layer heatmap for Gemma 3 4B. Four panels showing δ -cosine stability across generation steps.

Figure A3: Per-layer weight-change localization (PT $\rightarrow$ IT) across the five dense families plus a separate DeepSeek MoE side case. Gemma has large absolute late-window weight changes but also strong mid-window changes; the other dense families show more diffuse profiles.

Figure A4: δ -cosine profiles across the five dense families plus a separate DeepSeek MoE side case. IT (solid) vs PT (dashed). Gemma shows the largest late IT-vs-PT shift; Llama shows the weakest sustained shift because its PT variant already has a substantial late δ -cosine profile.

Table A1: depth-ablation effect normalized by window weight-change proxy. Δ_{KL} is the matched-prefix PT-side final-20% B_window - A' effect. Mean ΔW is the mean per-layer MLP weight-change proxy from Figure A3 over the same graft window. $\Delta_{KL} / \text{Mean } \Delta W$ is a descriptive scale-normalized diagnostic, not a formal parameter-efficiency estimand. Late is the largest raw effect and the largest normalized effect in every family, while the largest weight-change window is not consistently late.

Family	Early $\Delta_{KL} / \text{Mean } \Delta W$	Mid $\Delta_{KL} / \text{Mean } \Delta W$	Late $\Delta_{KL} / \text{Mean } \Delta W$	Largest ΔW window
Gemma 3 4B	-5.5	-14.0	146.6	Mid
Llama 3.1 8B	-236.7	-148.9	453.2	Late
Qwen 3 4B	-38.8	-60.1	184.4	Late
Mistral 7B	878.2	736.2	1754.4	Mid
OLMo 2 7B	0.6	27.1	130.3	Mid
DeepSeek-V2-Lite	-11.4	-55.1	234.7	Late

The dense-family mean raw effect is +0.341 nats late versus -0.035 early and -0.045 mid, while mean window weight-change is nearly identical for middle and late (0.00180 vs 0.00179). Thus the late intervention effect is not explained by a systematically larger late MLP weight delta in the dense-family pool.

Table A2: per-family raw late effect and leave-one-family-out sensitivity. Mistral is extreme only on the weight-normalized ratio because its Mean ΔW denominator is unusually small; it is not an outlier in raw late effect and does not drive the dense-family mean.

Family	Raw late ΔKL	Late Mean ΔW	Late ΔKL / Mean ΔW	Dense-4 mean if removed
Gemma 3 4B	+0.609	0.004156	146.6	+0.274
Llama 3.1 8B	+0.310	0.000685	453.2	+0.349
Qwen 3 4B	+0.491	0.002661	184.4	+0.304
Mistral 7B	+0.115	0.000066	1754.4	+0.398
OLMo 2 7B	+0.181	0.001392	130.3	+0.381
Dense-5 mean	+0.341	—	—	—

The leave-one-family-out means remain positive in every case. Removing Mistral increases the dense-family mean, so the pooled raw late-graft effect is not Mistral-driven. The large Mistral normalized value instead reflects a small absolute PT→IT MLP RMS weight-change denominator ($6.6e-5$) paired with a modest positive raw effect.

Figure A5: Cross-model δ -cosine heatmaps. Full layer x generation-step heatmaps for the five dense families plus the DeepSeek side case (PT and IT side by side), showing how the δ -cosine diagnostic varies across layers and generated positions.

A.2 Feature-level supplements

Figure A6: Feature importance analysis. Per-feature contribution to late post-training computation at layers 20–33, showing the distribution of importance across transcoder features.

Figure A7: Feature population dynamics. Gini coefficient and N50 distributions for IT vs PT at late layers, quantifying the broadening of the active feature repertoire.

A.3 Probe and commitment robustness

This subsection supports the layerwise context claim: the delayed-stabilization pattern is visible under raw and tuned probes, survives endpoint matching, and is not a Gemma-only tuned-lens artifact. These figures explain why late windows were worth testing, but they are not the paper's primary intervention estimand.

Figure A8: Tuned-lens validation. $KL(layer\ ell || final)$ for the five dense PT variants plus the DeepSeek side case. Red = tuned logit lens, blue = raw logit lens. The tuned lens substantially reduces KL at intermediate layers for Llama, Qwen, Mistral, and OLMo, with the DeepSeek side case behaving similarly. Gemma improves only modestly at comparable depth, indicating small tuned-vs-raw improvement rather than total probe failure. We therefore report both tuned and raw results throughout, and interpret Gemma's tuned-lens thresholded metrics with extra caution.

Table A3: raw-vs-tuned sensitivity for the convergence-gap context metric. Values are mean IT-minus-PT $KL(layer || own\ final)$ differences from the existing cross-family layerwise summaries. The raw lens does not remove the dense-family effect: the dense-5 final-half convergence gap is larger under raw lens (0.771) than tuned lens (0.410), and Gemma's own raw final-half gap (1.008) is larger than its tuned value (0.351). Excluding Gemma leaves the dense-family late-half raw gap positive (0.712), so the dense-family late stabilization result is not driven by Gemma's weak tuned-lens probe. The all-run row with DeepSeek is reported only as an appendix side case. This table is a sensitivity check for the layerwise readout, not a raw-only rerun of every matched-prefix KL intervention; the matched-prefix JS replay and first-divergence token projections are the non-tuned companion evidence for those later claims.

Scope	Lens	Early third	Middle third	Late third	Final-half CG	Full-stack mean
Dense-5 + MoE side case	Tuned	0.617	0.558	0.303	0.398	0.492
Dense-5 + MoE side case	Raw	0.330	0.598	0.638	0.729	0.526
Dense-5	Tuned	0.616	0.536	0.329	0.410	0.493
Dense-5	Raw	0.177	0.420	0.771	0.771	0.461
Dense-5 excluding Gemma	Tuned	0.487	0.545	0.324	0.425	0.453
Dense-5 excluding Gemma	Raw	-0.228	0.217	0.734	0.712	0.251

Scope	Lens	Early third	Middle third	Late third	Final-half CG	Full-stack mean
Gemma only	Tuned	1.133	0.500	0.350	0.351	0.652
Gemma only	Raw	1.797	1.231	0.922	1.008	1.305

Figure A9: KL-to-final trajectories in Gemma 3 4B. IT (solid) shows elevated KL-to-final at late layers (20–33), converging to the 0.1 nat threshold later than PT (dashed).

Figure A10: Mind-change analysis in Gemma 3 4B. Per-layer mind-change rates by token category. IT's late layers (20–33) show a sharp spike in mind-changes, with many targeting structural and discourse tokens.

Figure A11: Adjacent-layer KL divergence in Gemma 3 4B. IT (solid red) shows three discrete revision phases: early (layers 5–6), mid (15–17), and late (27–28), while PT (dashed blue) shows lower and more uniform prediction revision across layers.

Figure A12: Candidate reshuffling in Gemma 3 4B. Number of unique top-1 candidates encountered up to each layer. IT (red) shows rapid expansion in late layers; PT (blue) stabilizes earlier.

Figure A13: Alignment tax localization in Gemma 3 4B. Fraction of total activation mass allocated to IT-amplified features by layer depth. Late layers (20–33) show 14–16% of activation mass at layers 28–33.

Figure A14: Raw vs tuned logit lens commitment scatter. Per-step top-1 commitment layer under raw (x-axis) vs tuned (y-axis) logit lens. Points below the diagonal indicate tuned lens commits earlier (i.e., the tuned lens reveals earlier convergence that the raw lens misses). For most models, the tuned lens detects commitment at earlier absolute layers — consistent with its more faithful intermediate predictions — while preserving the IT > PT ordering.

Figure A15: Alternative commitment definitions. Commitment delay under majority-vote ($\geq 90\%$ subsequent layers KL < 0.1) for tuned and raw logit lens. The delay pattern replicates under this more conservative definition.

Figure A16: KL threshold sensitivity (full). Mean commitment vs KL threshold τ for both tuned (red) and raw (blue) lenses. The IT–PT gap is consistent across thresholds from 0.05 to 1.0 nats.

Figure A17: Cosine and entropy commitment. Commitment defined via cosine similarity ($\cos(h_\ell, h_{\text{final}}) > 0.95$) and entropy convergence ($|H_\ell - H_{\text{final}}| < 0.2$). These representation-space metrics show minimal IT–PT difference, indicating that the convergence gap is mainly a logit-space phenomenon under these probes.

Figure A18: Commitment CDF by normalized depth. Cumulative distribution of commitment layers for PT (dashed) and IT (solid), four methods. The rightward shift of IT CDFs is visible under KL-based metrics but absent under top-1 for some models — confirming the delay is distributional, not merely an argmax effect.

Figure A19: Endpoint-matched convergence-gap check. Token steps are matched within `model x probe_family` on final-layer entropy, final top-1 confidence, and final top-1/top-2 margin. IT retains a higher late `KL(layer || own final)` under raw and tuned probes, and endpoint-free path metrics also remain positive.

Table A4: endpoint-matched convergence-gap control. Dense-family endpoint-control run over 600 prompts per PT/IT branch. Matching is coarsened exact matching within `model x probe_family` on final entropy, final confidence, and final top-1/top-2 margin.

Quantity	Estimate		
Raw-probe late 'KL(layer		own final)', IT - PT	+0.425 nats, 95% CI [+0.356, +0.493]
Tuned-probe late 'KL(layer		own final)', IT - PT	+0.762 nats, 95% CI [+0.709, +0.814]
Remaining adjacent JS after endpoint matching, IT - PT	+0.052, 95% CI [+0.048, +0.057]		
Future top-1 flips after endpoint matching, IT - PT	+0.203, 95% CI [+0.190, +0.215]		
Minimum matched-token retention across model/probe branches	0.796		
Maximum post-match endpoint-covariate SMD	0.057		
Maximum malformed branch rate	0.000		

A.4 Matched-prefix localization and diagnostic behavioral figures

This subsection supports the late-window localization claim under identical token histories. The load-bearing artifacts are the graft/swap depth ablations and the matched random late-MLP KL control; the free-running behavior figures are consequence checks, not localization evidence.

Figure A20: Matched-prefix MLP graft trajectories across the five dense families plus a separate DeepSeek-V2-Lite MoE case. The intact IT model generates freely; the PT teacher-forced control and grafted PT branches are then forced to follow the same continuation. Solid lines show the raw-prompt branch and dashed lines the chat-template branch. The graft consistently reduces cross-KL to the IT teacher while reproducing auxiliary layerwise diagnostics only partially in the dense-model pool.

Figure A21: PT-side graft depth ablation. Equal-width early, middle, and late IT-MLP grafts are compared under identical teacher-forced token histories in a PT host. The late graft is the only window that consistently induces the final-window convergence-gap increase across the five dense families.

Figure A22: Free-running A/B/C output evaluation overview across all four judged metrics. A = PT raw, B = PT + late IT MLP graft under the same raw prompt, C = full IT model with its native chat template. B moves consistently toward C on benign false-refusal reduction, more selectively on assistant register, and only weakly on broad structure and harmful-prompt refusal.

Figure A23: Improvement relative to the PT baseline in the free-running A/B/C evaluation. Red = $B - A$, green = $C - A$. Positive bars indicate improvement for G1, G2, and S1; for S2, positive bars indicate a reduction in false refusal. The graft consistently captures part of the A→C gap, but remains well short of the full IT endpoint on most metrics.

Figure A24: Cross-family descriptive token-type analysis of the matched-prefix late stage. Left: displaced vs supported token classes under $A' \rightarrow B_{\text{late}}$. Center: teacher-token rank gain by collapsed token type. Right: token-type rank gain under early, middle, and late graft windows on the subset with recoverable raw depth traces. The late stage broadly supports the eventual teacher token and suppresses FUNCTION/OTHER raw-continuation-style alternatives, with a secondary formatting/discourse component.

Figure A25: Descriptive token-support appendix view. Per-model panels, candidate entry/exit distributions, and mind-change summaries for the matched-prefix token-type analysis.

Figure A26: Symmetric matched-prefix graft/swap summary. Left: PT-side late-region KL deltas for early, middle, and late IT-MLP grafts relative to A' . Center: IT-side late-region KL deltas for early, middle, and late PT-MLP swaps relative to C. Right: dense-family predictive correlations for output-relevant late-stage summaries (support_teacher, anti_top1, anti_kl_final) and δ -cosine.

Figure A27: Symmetric graft/swap appendix view. Per-model bidirectional window-effect panels and late-stage diagnostic summaries for the matched-prefix graft/swap analysis.

Figure A28: Matched random-control specificity check. Actual IT graft deltas are compared with matched random residual-projection controls on final-20% KL-to-own-final. The dense-family late true effect is large while the matched random late effect is near zero, ruling out a generic same-window perturbation account for the main late KL result.

Figure A29: Assistant-facing bucket deltas in the free-running LLM-judge behavioral follow-up. Dense-family pooled G2 deltas by prompt bucket for PT-side grafts and IT-side swaps, highlighting that the largest late judge-rated degradation on the IT side concentrates on conversational and register-sensitive prompts.

A.5 Same-history JS and candidate/amplification details

This subsection supports the identity/margin decomposition. Same-history JS shows that PT/IT distributions differ under matched histories; first-divergence token-transfer figures show the relative middle-over-late identity effect; margin-flow figures show why late windows still matter for readout.

Figure A30: Matched-prefix native same-layer JS divergence under identical teacher tokens. Per-model $JS(A', C)$ curves and dense-family pooled summaries show that broad PT↔IT output divergence is already present through much of the stack and amplifies late even when teacher histories are frozen.

Figure A31: Matched-prefix JS control view. PT-side and IT-side target-gap closure bars and host-local perturbation controls under matched prefix show that direct same-layer gap closure is more mid-to-late distributed than purely late, motivating the paper's "broad circuit, late delayed-stabilization window" synthesis rather than a strictly late-confined story.

Figure A32: Reverse teacher-stream JS check. Replaying the same matched-prefix native-JS analysis with PT-generated continuations as teacher tokens again shows a broad dense-family PT $\leftrightarrow$ IT same-layer JS gap. Late amplification is teacher-stream-dependent under token-step weighting, while prompt-mean aggregation still rises late; this supports the broad identical-history divergence claim while arguing against a strict teacher-stream-invariant late-confined interpretation. The Llama reverse replay excludes 11 empty PT-teacher continuations.

Figure A33: Token identity at the first PT/IT divergent token. Under the raw-shared prompt, middle swaps transfer opposite-model token identity more than late swaps, while native prompting shows the deployment-format counterpart.

Figure A34: Mid-vs-late IT-token margin effects. Late windows dominate IT-vs-PT token-margin changes, especially under the native IT chat template.

Figure A35: Raw-shared per-model token-transfer heatmap. Across dense families, middle swaps generally transfer token identity more than late swaps, with DeepSeek reported separately as the MoE case in all-model outputs.

Figure A36: Native PT/IT final-token margin flow. Dense-5 pooled mid and late window margin deltas for the finally emitted token, plus dense-family mid-selected/late-helped rates and per-model IT FORMAT late-minus-mid margin. Late IT windows produce the strongest margin gains, especially for FORMAT and CONTENT, while PT is flatter.

Figure A37: Matched-prefix candidate/amplification chronology. Left: teacher-token rank gain by disjoint window under identical histories. Middle: strict rate at which a token is first selected in the middle window and then further helped late. Right: continuity view of $A' \rightarrow B_window$ top-1 displacement. Format-like tokens become teacher-rank-positive only late, while content shows larger middle-window gains.

A.6 DeepSeek/MoE side case

DeepSeek-V2-Lite is retained only as a descriptive side case wherever the experiments were run. It is not pooled into the main dense-family intervention, behavioral, endpoint-control, or first-divergence claims. We did not add a second MoE model or analyze router/expert-selection mechanisms, so the paper makes no claim about MoE generalization. The reason is methodological rather than cosmetic: in a MoE checkpoint, an MLP graft can change both expert computation and routing/expert selection, which is not directly comparable to dense MLP substitutions without additional controls.

A.7 Qwen2.5-32B dense-family member

Exp24 repeats the core first-divergence battery on Qwen/Qwen2.5-32B and Qwen/Qwen2.5-32B-Instruct. This is the sixth dense PT/IT pair in the Dense-6 core factorial estimate; it is still only one 32B family, not a pooled 32B-scale claim. The run covers Exp20 identity/margin, Exp21 MLP write-out, Exp23 residual-state $\times$ late-stack factorial, raw-KL bridge, and a 32B-specific first-divergence position audit. It preserves the central sign: the residual-state $\times$ late-stack interaction is positive under both common readouts, while the simple PT-upstream late effect is smaller than the IT-upstream late effect.

Qwen2.5-32B readout	Estimate	95% CI	Interpretation
Exp23 late IT given PT upstream	+0.977	[+0.880, +1.075]	Late stack helps from PT upstream, but less.
Exp23 late IT given IT upstream	+2.423	[+2.265, +2.583]	Same late stack helps more from IT-shaped upstream.
Exp23 upstream $\times$ late interaction, common-IT readout	+1.446	[+1.321, +1.569]	Sixth dense-family contribution to the Dense-6 main estimand.
Exp23 upstream $\times$ late interaction, common-PT readout	+1.478	[+1.346, +1.606]	Readout-choice cross-check.
Exp23 position ≥ 3 interaction, common-IT readout	+1.020	[+0.853, +1.205]	Not only first-response formation.
Exp23 position ≥ 5 interaction, common-IT readout	+0.686	[+0.521, +0.875]	Later-position signal remains positive.
Exp23 position ≥ 10 interaction, common-IT readout	+0.886	[+0.589, +1.223]	Thin but positive later subset.
Exp20 PT-host IT-token transfer, middle vs late	45.7% vs 27.4%	—	Middle substitutions are more identity-moving.
Exp21 raw-shared late MLP margin write-in	+0.0498	[+0.0457, +0.0536]	Token-specific late write-out persists.
Raw-KL bridge, IT-side final-20% interaction	+0.465 nats	[+0.435, +0.494]	IT-side convergence-gap interaction persists.
Raw-KL bridge, PT-side final-20% interaction	-0.125 nats	[-0.133, -0.118]	Not a symmetric KL-positive claim.

The Qwen2.5-32B first-divergence distribution is later and more mixed than the five-family 4B-8B holdout: 1,397 valid prompt clusters, mean generated position 4.66, median 2, and 628 clusters at position ≥ 3 . Prompt categories are not a single residue (CONTENT-FACT 21.5%, GOV-CONV 21.5%, GOV-FORMAT 17.8%, CONTENT-REASON 14.2%, plus smaller safety/register/baseline groups). At position ≥ 3 , the interaction is positive for content factual (+1.47, 95% CI [+1.05, +1.93]) and content reasoning (+1.58, 95% CI [+0.90, +2.33]) subsets. These rows characterize the 32B family contribution to the Dense-6 result, not a separate pooled 32B-scale result.

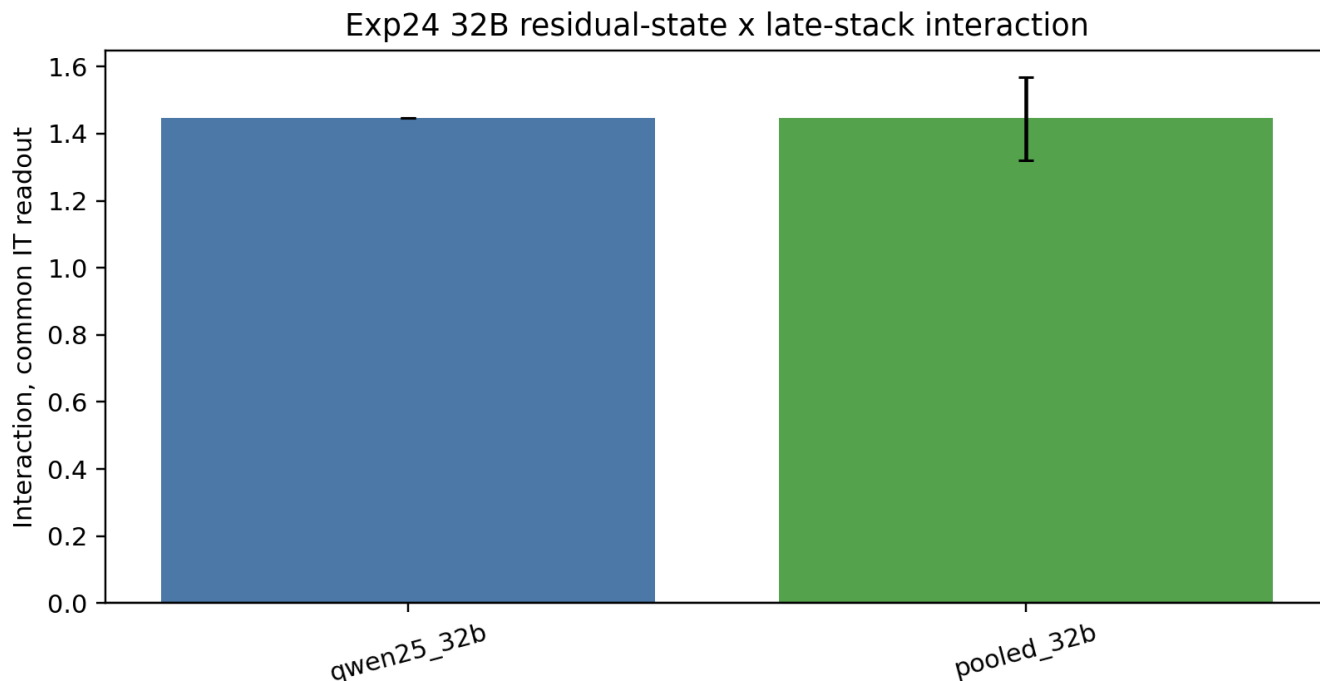


Figure A38: Qwen2.5-32B dense-family member. The residual-state x late-stack interaction remains positive in the 32B pair and supplies the sixth family in the Dense-6 core estimate.

A.8 OLMo-2 stage-progression case study

Exp25 uses the released OLMo-2 7B Base, SFT, DPO, and Instruct/RLVR checkpoints to ask whether the upstream $\times$ late interaction appears in adjacent post-training transitions, not only in the full Base $\rightarrow$ Instruct/RLVR contrast. This is a single-lineage case study and is not pooled with the Dense-6 core estimate. The key caution is that each adjacent transition has its own first-divergence events and divergent token pair, so the adjacent estimates are local transition measurements rather than additive components of the full Base $\rightarrow$ RLVR number.

Table A5: OLMo-2 7B local transition estimates. All rows use raw-shared first-divergence factorial diffing under the common later-stage readout. The "later-stage late effect" and "earlier-stage late effect" columns are the simple late-stack effects from later-shaped and earlier-shaped upstream states, respectively.

Transition	Interaction (95% CI)	Later-stage late effect	Earlier-stage late effect	IT-token transfer proxy	Interpretation
Base $\rightarrow$ SFT	+0.782 [+0.681, +0.882]	+1.328	+0.546	mid+late 40.8% vs mid 24.1%	Strongest adjacent-stage interaction.
SFT $\rightarrow$ DPO	+0.135 [+0.114, +0.157]	+0.821	+0.686	mid+late 57.1% vs mid 30.7%	Smaller preference-stage refinement.
DPO $\rightarrow$ RLVR	+0.016 [+0.008, +0.025]	+0.098	+0.082	mid+late 43.8% vs mid 35.0%	Positive but tiny local addition.
Base $\rightarrow$ RLVR	+1.930 [+1.749, +2.112]	+2.698	+0.768	mid+late 36.0% vs mid 18.4%	Full contrast remains large.

The stage-progression result fits the paper's main reading: post-training creates or amplifies a compatibility relation between upstream state and late computation, with the largest local OLMo jump appearing at SFT and a much smaller additional RLVR-stage change under this readout. Exp20 collection completed all four transitions at 600/600 records; valid first-divergence

units were 551 for Base→SFT, 557 for SFT→DPO, 466 for DPO→RLVR, and 586 for Base→RLVR. Preflight checks found no raw-prompt token-ID mismatches across the stage pairs; all variants use GPT2TokenizerFast, shared vocab size 100278, EOS <|endoftext|> (100257), and pad <|pad|> (100277). The SFT, DPO, and Instruct/RLVR tokenizers share the same chat-template hash, while the raw-shared factorial uses raw text for both branches.

Exp25 OLMo 2 7B stage progression: local transition estimates

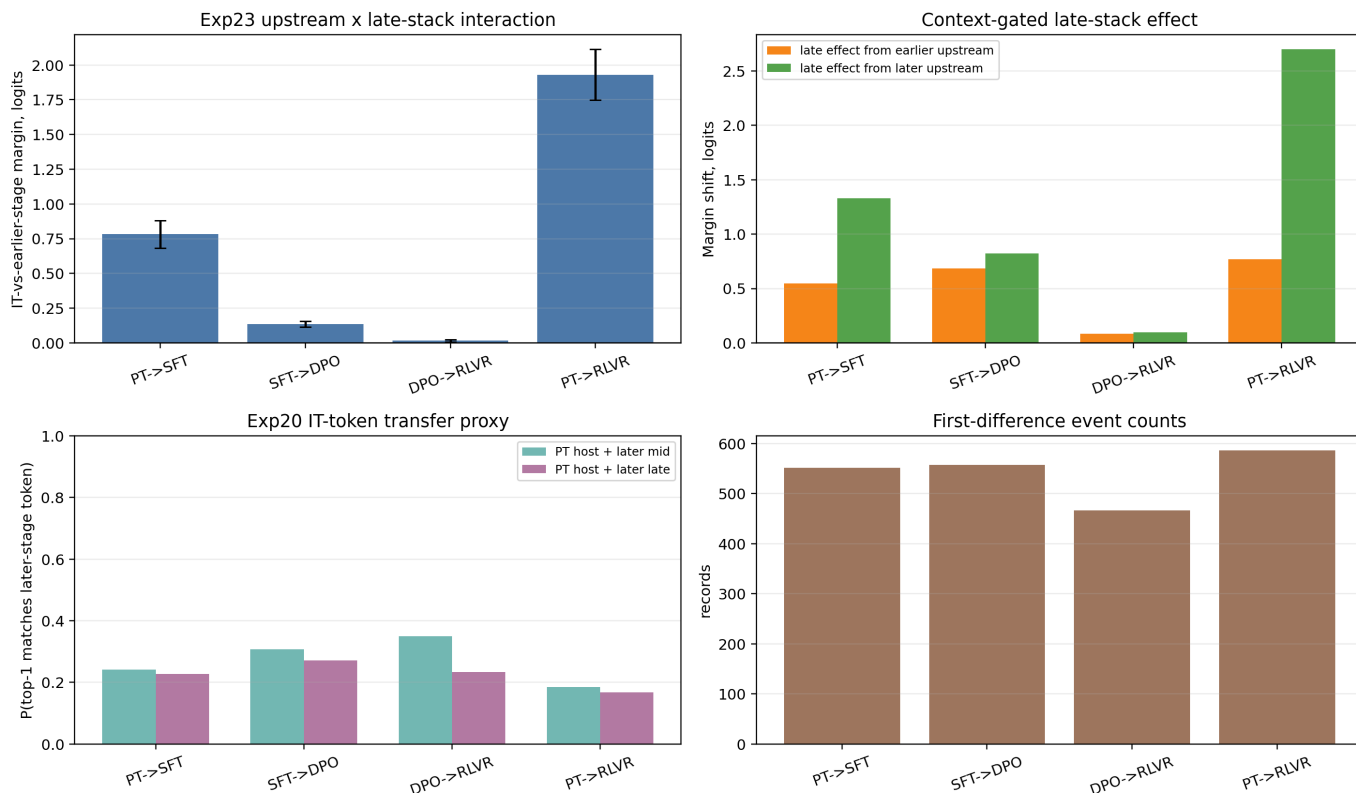


Figure A39: OLMo-2 7B stage progression. Local first-divergence transition estimates show positive upstream x late interaction at each released stage transition, with the strongest adjacent transition at Base→SFT and a large full Base→RLVR contrast.

A.9 Auxiliary mechanism check: natural-rollout residual opposition

Exp27 asks a simpler question than the first-divergence factorial: during deployment-style greedy generation, how much does each model rely on its own late residual-opposing MLP component to predict the tokens it actually emits? For each PT and IT checkpoint in the Dense-5 support set, we generate a native greedy continuation from the same prompt set (PT raw prompt; IT native chat format), then replay that model's own generated continuation under full and ablated late-MLP variants. The intervention is applied only on generated-token source positions. Positive NLL hurt means the ablation worsens prediction of the model's own generated tokens; true-logit drop is the corresponding drop in the logit of the generated token.

This is an auxiliary natural-rollout importance measurement, not a same-prefix PT/IT causal comparison and not a mediation test for the upstream × late interaction. Its value is that it checks whether the residual-opposing component is consequential in normal IT generation. It is: removing the component barely changes PT own-token NLL on average but consistently hurts IT own-token prediction, and the direct true-logit drop is much larger for IT than PT. Norm-preserving removal keeps the conclusion, while same-magnitude random removal hurts PT more in NLL, so the result is not simply that IT is more fragile to any late-vector deletion.

Exp27 natural-rollout residual-opposition ablations

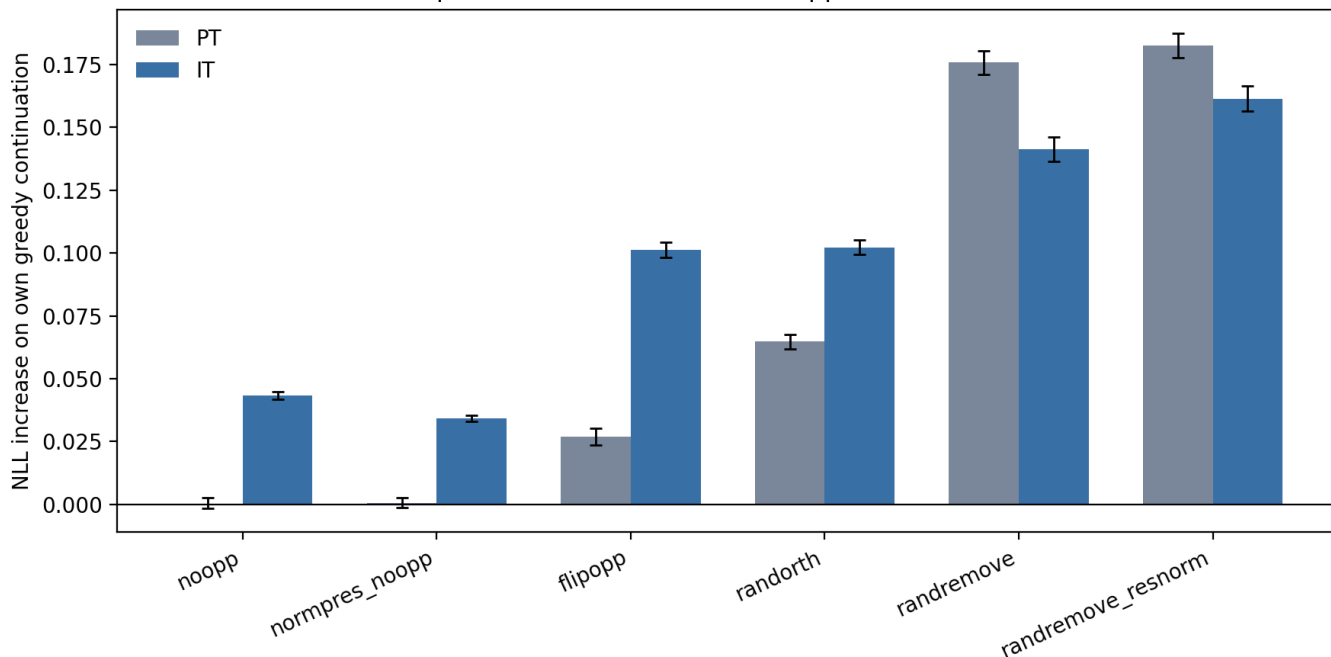


Figure A40: Natural-rollout residual-opposition importance. Late residual-opposing MLP ablations hurt each model's own generated-token prediction much more in IT than PT for structured no-opposition and sign-flip interventions; same-magnitude random-removal controls show this is not a generic late-vector fragility effect.

Table A6: natural-rollout residual-opposition importance. Dense-5 run over 600 prompts per PT/IT family. The analyzed table contains 372, 144 generated-token source-position units after aggregating random seeds. IT - PT compares each model's own native rollout, so it should be read as deployment-style importance rather than a matched-history causal effect.

Variant	PT NLL hurt	IT NLL hurt	IT-PT NLL hurt	PT true-logit drop	IT true-logit drop	IT-PT true-logit drop	Interpretation
Remove opposition (noopp)	+0.0004 [-0.0016, +0.0027]	+0.0432 [+0.0418, +0.0448]	+0.0428 [+0.0403, +0.0453]	+0.83	+7.37	+6.54 [+6.40, +6.68]	Structured removal matters much more for IT own-token prediction.
Norm-preserving removal	+0.0007 [-0.0012, +0.0027]	+0.0342 [+0.0329, +0.0356]	+0.0336 [+0.0312, +0.0357]	+0.51	+6.35	+5.84 [+5.70, +5.97]	Not explained by shrinking MLP-update norm.
Flip opposition	+0.0269 [+0.0237, +0.0304]	+0.1013 [+0.0983, +0.1043]	+0.0744 [+0.0699, +0.0786]	+1.73	+10.91	+9.19 [+9.01, +9.36]	The opposing direction's sign is important for IT.
Same-magnitude random removal	+0.1758 [+0.1710, +0.1804]	+0.1414 [+0.1364, +0.1461]	-0.0344 [-0.0397, -0.0293]	+1.43	+4.40	+2.97 [+2.91, +3.03]	Generic same-size deletion is not the explanation.
Random removal, residual-norm matched	+0.1826 [+0.1778, +0.1874]	+0.1614 [+0.1564, +0.1664]	-0.0213 [-0.0267, -0.0160]	+1.81	+9.02	+7.22 [+7.07, +7.36]	Residual-norm matching does not erase the IT true-logit dependence.

The strongest defensible conclusion is absolute importance, especially for direct own-token logit support. We do not claim from this table that IT has universally larger per-unit NLL sensitivity to residual opposition, nor that residual opposition alone mediates the first-divergence upstream $\times$ late interaction. Instead, Exp27 supports the auxiliary mechanism picture: post-training makes this late residual-opposing component much more consequential for the IT model's own next-token readout, while the main paper's causal estimand remains the first-divergence upstream $\times$ late interaction.

Appendix B: Auxiliary Consequence Check: Evaluation Methodology

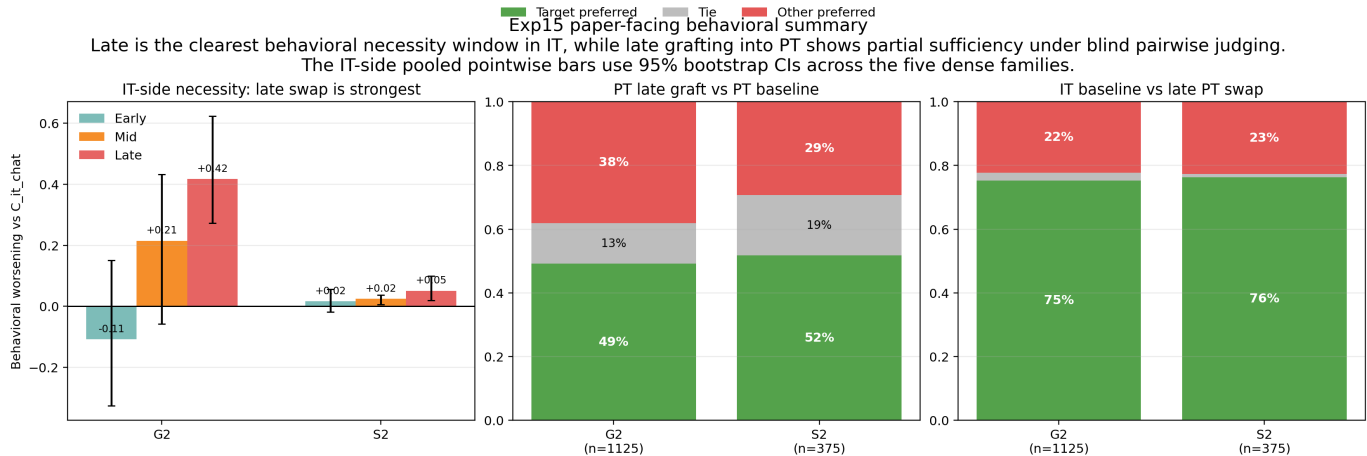


Figure B1: Diagnostic free-running behavioral check. The same late intervention family produces measurable LLM-judge behavioral movement under natural decoding, but the figure is appendix material because behavior is an aggregate consistency check rather than core internal localization evidence.

B.1 LLM judge design and rubric definitions

The behavioral run uses four pointwise judge tasks and two blind pairwise judge tasks. G1 is a 1-5 score for response structure and formatting, independent of factual correctness. G2 is a 1-5 score for assistant-like conversational register versus raw web-text continuation, applied to assistant-facing prompts. S1 classifies harmful-request responses as REFUSE, COMPLY, or INCOHERENT; incoherence is not counted as refusal. S2 is a binary benign false-refusal metric, where 1 means the model refused, over-hedged, or failed to engage with a safe request. Pairwise G2 and S2 ask which of two anonymized responses better satisfies the same criterion, with randomized A/B ordering and a TIE option.

Pointwise judging uses a bulk model (google/gemini-2.5-flash) and a second model (openai/gpt-4o-mini) through the OpenRouter-compatible interface. The manifest also records an escalation model (openai/gpt-4o), parser versions, rubric hashes, dataset hash, pipeline hash, and output hash. Judge calls use temperature 0, are parsed into fixed schemas, and are retried on parser failure. Pointwise items are escalated if the two judge models differ by at least 2 points on G1/G2, disagree on S1, or disagree on S2. Pairwise judging uses the bulk model, is blind to model family and condition names, and uses randomized A/B presentation for B_late versus A, C versus D_late, and C versus A.

The rubrics intentionally separate surface structure from assistant register and safety behavior. G1 asks only about structure/formatting; G2 asks whether the response behaves like a helpful assistant addressing the user; S1 asks whether harmful prompts are refused; and S2 asks whether benign safety prompts are incorrectly refused or over-hedged. Pairwise G2 and S2 are the primary behavioral readouts because they are condition-blind, randomized, and closer to the human-preference setup used in LLM-as-judge work. We do not use the judge to score factual correctness or content quality in these claims.

Known LLM-judge failure modes motivate this design. Prior work finds that strong LLM judges can approximate human preferences but are sensitive to position, verbosity, self-enhancement, rating indeterminacy, and rubric design (Liu et al., 2023; Zheng et al., 2023; Wang et al., 2023; Dubois et al., 2024; Guerdan et al., 2025). We therefore use condition blinding, randomized order, a TIE option, fixed rubrics, schema-validated outputs, model-disagreement escalation, bootstrap uncertainty, and a completed blind human audit rather than treating the automated judge as ground truth.

B.2 Human audit results

The current behavioral run materialized blind human-audit packs for all five dense families: 120 pointwise audit items per model, 600 total. The primary pairwise human packet contains 1,200 blinded comparisons per rater: 60 items per dense model for each of four primary contrasts, C versus D_late on G2, C versus D_late on S2, B_late versus A on G2, and B_late versus A on S2. Four completed rater sheets (R1-R4) are present in paper_draft/human_eval_survey/pointwise/ and paper_draft/human_eval_survey/pairwise/; the hidden keys are kept separately under paper_draft/human_eval_survey/keys/.

The human protocol follows standard NLG human-evaluation practice: blinded rating, unchanged row IDs, no access to condition or model labels, confidence ratings, optional notes for ambiguity, and frozen labels before unblinding (van der Lee et al., 2019).

We do not adjudicate disagreements into a forced single label, because rating indeterminacy is itself informative for these outputs. Pairwise resolved rates therefore exclude TIE and BOTH_BAD, but those unresolved votes are reported explicitly. Confidence intervals bootstrap over `pair_id` clusters, with all completed rater votes included for each sampled item.

Contrast	Criterion	Human resolved target win (95% CI)	Resolved votes	Unresolved vote rate	LLM same-sample resolved target win
C over D_late	G2	73.1% [66.5, 79.5]	398/1200	66.8%	76.5%
C over D_late	S2	71.8% [61.8, 80.8]	181/1200	84.9%	75.4%
B_late over A	G2	60.2% [52.9, 67.5]	540/1200	55.0%	62.1%
B_late over A	S2	65.0% [57.4, 71.8]	531/1200	55.8%	65.7%

The audit agrees directionally with the LLM judge on all four primary pairwise contrasts, and all four pooled human confidence intervals are above chance on resolved votes. The result is stronger after the additional rater sheets, but should still be read conservatively. Ties and both-bad labels remain common, especially for IT-vs-D_late benign-safety comparisons, and some per-family cells have small resolved counts or flip direction. We include this appendix to answer the basic output-consequence sanity question; the behavioral audit is not part of the internal evidence spine and is not a per-family behavioral localization claim.

Contrast	Criterion	Mean pairwise raw agreement	Mean pairwise κ	Fleiss κ	Collapsed Fleiss κ
C vs D_late	G2	70.4%	0.40	0.40	0.40
C vs D_late	S2	82.3%	0.52	0.42	0.44
B_late vs A	G2	69.3%	0.54	0.54	0.62
B_late vs A	S2	71.1%	0.61	0.61	0.73

Mean pairwise κ averages Cohen's κ over rater pairs; Fleiss κ is the multi-rater nominal agreement statistic. Collapsed κ merges TIE and BOTH_BAD into a single unresolved label. Agreement is now moderate on all four pairwise contrasts, with the weakest reliability still on the IT-side assistant-register comparison. This motivates using the human audit as directional confirmation of the behavioral sanity check, not as a high-precision substitute for the automated judge.

Pointwise audit labels, on common/applicable filled fields, are used only as judge-calibration diagnostics. Mean pairwise rater κ is 0.78 for G1, 0.65 for G2, 0.72 for S1, and 0.68 for S2. Human-vs-judge κ is 0.62 for G1, 0.48 for G2, 0.61 for S1, and 0.20 for S2. These values support the aggregate direction of the behavioral check, but the high unresolved rate in the pairwise audit is why behavior remains a sanity check rather than a mechanism-identification endpoint.

B.3 Statistical testing approach

Pointwise behavioral summaries are prompt-bootstrap estimates over dense-family records. Pairwise summaries report target win rate, other win rate, tie rate, and resolved win rate, with bootstrap 95% confidence intervals. Dense-family claims pool Gemma, Llama, Qwen, Mistral, and OLMo; DeepSeek is excluded from the canonical behavioral pool because its MoE routing makes dense-family MLP graft/swap interpretation non-comparable. The main LLM-judge behavioral claim is deliberately asymmetric: IT-side late-window degradation is strong, while PT-side late grafting produces partial movement mainly supported by blind pairwise preferences.

For the completed human audit, the primary diagnostic quantities are pairwise resolved win rates and bootstrap confidence intervals for the same four primary contrasts shown in Figure B1. The rater packets randomize A/B order, expose no model or condition labels, and include TIE/BOTH_BAD to avoid forcing noisy preferences. The human audit was analyzed after the automated judge setup, endpoint definitions, and primary contrasts were fixed, so it functions as an external validation check rather than as a source of endpoint selection. We report same-sample LLM rates beside the human rates to show direction agreement directly.

B.4 Prompt dataset construction

The behavioral run uses a frozen 600-prompt core subset of `eval_dataset_v2.jsonl`. The subset emphasizes conversational and register-sensitive prompts, benign safety prompts, harmful safety prompts, and format-sensitive items. Pointwise G1 applies broadly; G2 applies only to assistant-facing records; S1 applies to harmful safety records; and S2 applies to benign safety records. The judge manifest stores hashes for the dataset manifest, pipeline manifest, human-audit manifest, and sample outputs so reruns can detect accidental dataset or pipeline drift.

Appendix C: Token Classifier Specification and Robustness

C.1 Classifier specification

Five categories with priority order (first match wins): STRUCTURAL (regex-matched, 2.8% baseline), PUNCTUATION (1.2%), DISCOURSE (Hyland 2005 taxonomy, 0.5%), FUNCTION (closed-class, 34.9%), CONTENT (default, 59.9%).

C.2 Perturbation robustness analysis

Four perturbation scenarios tested. Only the STRUCTURAL/DISCOURSE boundary produces measurable sensitivity ($\Delta = 0.145$ when merged — expected, as both capture formatting). Content-side perturbations produce $\Delta < 0.00001$. No core finding depends on precise boundary choices.

Appendix D: Commitment Threshold Sensitivity

The convergence-gap claim does not depend on a single commitment threshold. The main text summarizes the threshold-free top-1 commitment view; Appendix Figures A15-A18 show majority-vote, KL-threshold, cosine, entropy, and normalized-depth CDF variants. Across KL thresholds from 0.05 to 1.0 nats, the IT-minus-PT delay remains qualitatively stable.

Representation-space commitment summaries are weaker, which is consistent with the paper’s narrower claim that the robust PT↔IT separation is in decoded next-token distributions rather than arbitrary residual-stream distances.

Appendix E: Broader Literature Positioning

Table E1 is an orientation map, not a proof of priority. To keep the comparison disciplined, it includes accepted conference papers or field-standard sources that are directly used by the argument, rather than every adjacent arXiv preprint. Each ingredient in the table already appears somewhere in the literature: PT/post-training comparison, cross-model or cross-checkpoint patching, layer localization, behavioral intervention, first-divergent-token metrics, and conceptual accounts of late computation reading upstream state. The novelty claim is narrower: in this paper those ingredients are assembled into a measurable upstream-state x late-stack interaction at the first natural PT/IT next-token disagreement, with matched-history controls, separate identity/margin readouts, a label-swap permutation null for the factorial, and a matched random residual-projection control for the late-MLP KL analysis. The coarse labels below only indicate which ingredients are directly part of each paper’s design (Yes), present in a narrower task-specific form (Partial), or outside that paper’s main design (No).

Paper	PT→post-trained descendants?	Cross-family paired checkpoints?	Identical-history internal comparison?	Symmetric depth-localized intervention?	First-divergence candidate/margin?	State x late-stack factorial?	Natural-decoding consequence test?
Lad et al. (2025)	No	No	No	No	No	No	No
Joshi et al. (2025)	No	No	No	No	No	No	No
Wu et al. (2024)	Yes	No	No	No	No	No	No
Du et al. (2025)	Yes	Yes	No	No	No	No	No
Li et al. (2025)	No	No	No	No	No	No	No
Jain et al. (2024)	Yes	Partial	No	Partial	No	No	Yes
Prakash et al. (2024)	Yes	No	No	Partial	No	No	No
Deiseroth et al. (2024)	Partial	No	Yes	No	No	No	No
Zhao, Ziser, and Cohen (2024)	Yes	No	No	No	No	No	No
Panigrahi et al. (2023)	No	No	No	Partial	No	No	No
Panickssery et al. (2024)	No	No	No	No	No	No	Yes
Ours	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Lad et al. (2025) and Joshi et al. (2025) are the closest phenomenon-level comparisons. Lad gives a generic vocabulary for mid-to-late inference stages: prediction ensembling followed by residual sharpening. Joshi studies late confidence calibration after decision certainty has emerged. We do not treat late sharpening, late calibration, or late-stage confidence correction as new

phenomena. The delayed-stabilization analysis is the aggregate background signature; the contribution is the paired PT/IT first-divergence analysis that asks which depth windows change the divergent token identity and how the final IT-vs-PT margin depends jointly on upstream state and late computation.

Du et al. (2025) overlaps with the paper's basic model-diffing premise: compare base and post-trained descendants and ask how knowledge, truthfulness, refusal, and confidence change internally. That overlap is real and useful. Our narrower addition is to condition on the first natural PT/IT next-token disagreement and run matched-history interventions at that exact prefix, so the readout is not only "post-training changes confidence/refusal" but "this upstream state and this late stack interact on this divergent-token margin."

Prakash et al. (2024) is the closest methodological precedent in this broader set: CMAP patches activations across related base/fine-tuned models to reveal improved mechanisms. Our use is different in scope and target: we apply a symmetric equal-width early/mid/late MLP graft/swap design across model families, and the localized target is not entity tracking but natural PT/IT next-token disagreement under matched-prefix control. Following the caution urged by activation-patching work such as Heimersheim and Nanda (2024), we describe these interventions as effects on measured readouts, not as complete mechanism recovery.

Zhao, Ziser, and Cohen (2024) and Panigrahi et al. (2023) are important layer-localization precedents. Zhao, Ziser, and Cohen study where task-oriented representations appear in instruction-tuned models, and Panigrahi et al. study task-specific skill localization in fine-tuned language models. These papers make upstream dependence plausible rather than surprising. The distinction is not that we are the first to see layer structure after instruction tuning, and the paper should not be read as another qualitative layer-stage taxonomy. The distinction is the estimand: we estimate a window-level upstream-state x late-stack interaction at the first natural PT/IT token disagreement, with matched histories, separate candidate-identity and margin outcomes, and a PT/IT label-swap null. That design directly tests whether the late-stage effect is portable across upstream states; the factual/reasoning sign flip shows why this is not a cosmetic strengthening of the older heuristic.

Panickssery et al. (2024), Li et al. (2025), and Jain et al. (2024) are the main reason we do not frame the contribution as finding an instruction-following activation vector or safety layer. Those papers show that activation-space interventions or layer-localized analyses can affect behavior, including safety-relevant behavior. The paper-level contribution here is that paired PT/IT next-token disagreement is decomposed with graft/swap interventions under identical histories and first-divergence counterfactuals, rather than by extracting a behavioral vector.

The table should therefore be read as a map of overlap, not as a claim that novelty follows from a unique combination of checkmarks. The stronger claim is the estimand: prior ingredient-papers motivate pieces of the design, but they do not report an upstream-state x late-stack interaction on the actual first-divergent PT/IT token margin with the label-swap factorial control and matched random late-MLP KL control used here. Our paper asks a narrower and more operational question: what paired PT $\leftrightarrow$ post-training forward-pass signature appears across dense families, how middle and late windows divide candidate selection from margin effects, and whether that same intervention family matters under natural decoding.

Appendix F: Additional Scope Notes

The main limitations are stated in Section 5. This appendix keeps only the scope details that are useful for interpreting or reproducing the experiments.

The early/middle/late labels are coarse intervention windows, not discovered sharp phase boundaries. We use overlapping windows at comparable normalized depths so that the comparisons are not forced by a single brittle cutoff:

Family	Early window	Middle window	Late window
Gemma 3 4B	0 - 13	10 - 23	20 - 33
Llama 3.1 8B	0 - 12	9 - 21	19 - 31
Qwen 3 4B	0 - 13	11 - 24	22 - 35
Mistral 7B	0 - 12	9 - 21	19 - 31
OLMo 2 7B	0 - 12	9 - 21	19 - 31

Earlier boundary checks support this conservative reading. A Gemma layer-range sensitivity run varied the late intervention range across 18 - 33, 20 - 31, 20 - 33, and 22 - 33; broad late-range behavior survived, while magnitudes changed. An onset-threshold sensitivity run found that estimated onset layers move with threshold and family. We therefore interpret the interventions as depth-window tests rather than evidence for a single exact start layer.

Prompt formatting is part of the estimand rather than a nuisance to erase. Native layerwise runs use each checkpoint in its natural deployment regime: PT branches receive raw prompts and IT branches receive the checkpoint chat template. Raw-shared first-divergence controls force identical raw prompt token IDs before comparing residual states, and they preserve the qualitative middle-identity/late-margin split. We therefore read templates as part of native IT operation, not as the sole source of the effect.

First-divergence position is an informative axis of the phenomenon, so we stratify the Exp23 interaction by generated position. In the Dense-6 synthesis, the interaction remains positive when immediate first-token divergences are removed and in later shared-history regimes:

Position stratum	Records	Dense-6 interaction, logits (95% CI)	Gemma-removed interaction, logits (95% CI)
all positions	4,380	+2.44 [+2.35, +2.52]	+1.71 [+1.64, +1.78]
positions >=1	2,339	+2.08 [+1.96, +2.19]	+1.16 [+1.08, +1.24]
positions >=3	1,428	+1.43 [+1.30, +1.57]	+0.83 [+0.75, +0.91]
position >=5	922	+1.48 [+1.28, +1.68]	+0.80 [+0.70, +0.90]
position >=10	352	+1.39 [+0.82, +1.97]	+1.01 [+0.81, +1.21]

The more detailed five-family raw-record table below preserves the original position bins used by the category-mix audit:

Position stratum	Records	Dense-5 interaction, logits (95% CI)	Gemma-removed interaction, logits (95% CI)
all positions	2,983	+2.64 [+2.54, +2.73]	+1.77 [+1.69, +1.86]
position 0	1,499	+3.01 [+2.88, +3.16]	+2.38 [+2.25, +2.52]
positions 1-4	989	+2.42 [+2.26, +2.58]	+1.29 [+1.17, +1.41]
positions >=1	1,484	+2.25 [+2.11, +2.38]	+1.14 [+1.05, +1.23]
positions >=2	1,123	+1.77 [+1.63, +1.93]	+1.06 [+0.97, +1.15]
positions >=3	800	+1.52 [+1.36, +1.68]	+0.79 [+0.70, +0.88]
positions >=4	641	+1.52 [+1.33, +1.71]	+0.80 [+0.70, +0.91]
position >=5	495	+1.64 [+1.39, +1.88]	+0.83 [+0.71, +0.94]
position >=10	140	+1.49 [+0.85, +2.20]	+1.05 [+0.81, +1.29]

Per-family thresholding clarifies how the effect changes with first-divergence position and where the data become lower-powered. At generated position >=3, all five family-specific intervals are above zero, with Llama close to the boundary. At generated position >=5, the dense and Gemma-removed pooled estimates remain positive, but Llama's family-specific interval is compatible with zero (+0.116, 95% CI [-0.047, +0.269]). We therefore use >=3 for the stronger family-level later-position statement and treat >=5 and >=10 as lower-powered diagnostics of later shared-history disagreements.

A CPU-only category-mix audit checks what the >=3 threshold is selecting. It is composition-shifted, so it should be read as a later-position regime rather than a balanced prompt-category estimate: 800 records remain across all five dense families, with prompt categories GOV-CONV 700 (87.5%), GOV-FORMAT 61 (7.6%), and SAFETY 39 (4.9%). The divergent IT-token categories remain more mixed: CONTENT 476 (59.5%), FUNCTION_OTHER 199 (24.9%), and FORMAT 125 (15.6%).

Crossing position with prompt category gives the same dynamic read. GOV-CONV dominates the later-position subset, but its own interaction also changes from +2.05 logits over all positions to +1.51 logits at position >=3. GOV-FORMAT remains larger at position >=3 (+2.28 logits, 61 clusters), while SAFETY is positive but lower-powered (+0.64 logits, 39 clusters). Thus the later-position change is partly a category-composition effect and partly a within-category position effect.

Assistant-marker tokens are not the whole explanation. In the five-family holdout, 551/2,983 events contain an assistant-marker token; the non-assistant-marker stratum remains strongly positive (+2.48 logits, 95% CI [+2.38, +2.59]), while assistant-marker events are larger (+3.30, [+3.07, +3.53]). This is the pattern expected if response-opening conventions are one high-magnitude case of the same upstream-late compatibility effect, not the sole source of it.

The main convergence metric, `KL(layer || own final)`, is intentionally endpoint-relative: it measures stabilization toward a model's own eventual next-token distribution. Endpoint-matched controls balance final entropy, final top-1 confidence, and final top-1/top-2 margin, and matched-prefix JS gives a separate endpoint-free check. The first-divergence factorial remains the primary intervention estimand; the KL and JS analyses provide layerwise context.

First-divergence intervals use prompt-cluster bootstraps within fixed families. Appendix G gives the exact clustering rule, including the content/reasoning extension where multiple event records can share one prompt.

The remaining limitation is circuit granularity. The evidence supports window-level intervention leverage and upstream-late non-additivity in constructed hybrid forward passes, not a named circuit with identified features, heads, and subspaces. The natural next tests are feature- or subspace-level mediation inside the middle-to-late handoff, larger dense checkpoints, and a non-dense design that separates MoE routing from expert computation.

Appendix G: Reproducibility and Artifact Map

Public release. During double-blind review, the project repository is provided through an anonymized artifact archive. It includes the current manuscript draft, all source experiment packages under `src/poc/`, shared cross-model infrastructure under `src/poc/cross_model/`, grouped script entrypoints under `scripts/`, and committed paper-facing artifacts under `results/`. The canonical prompt and evaluation datasets are committed under `data/`, including `eval_dataset_v2.jsonl`, `eval_dataset_v2_holdout_0600_1199.jsonl`, `exp3_dataset.jsonl`, `exp6_dataset.jsonl`, and `gold_standard_v1.csv`.

Intervention and uncertainty details. The first-divergence residual-state patch replaces the full hidden-state sequence entering the late-boundary transformer block, not only the final-token vector. Donor and host branches share raw prompt IDs, generated-prefix IDs, and attention masks; downstream attention and MLP computations are recomputed with `use_cache=False`, so K/V caches are not reused across patched cells. The diagonal factorial cells (`U_IT, L_IT`) and (`U_PT, L_PT`) are no-op reconstructions of native IT and PT forward passes. Runtime assertions require their hidden states to match the unmodified native hidden states within $1e-4$ max-absolute deviation, a bf16-safe tolerance whose observed deviations sit at the bf16 round-trip floor.

Unless otherwise stated, first-divergence factorial intervals are 95% percentile bootstraps over `model × prompt_id` clusters. When several event records share one prompt, the analysis first averages the effect within that prompt and then resamples prompt clusters within each dense family before averaging family estimates. The five-family 4B-8B holdout analyzes only `first_diff`, so its 2,983 valid records are exactly 2,983 prompt clusters; Qwen2.5-32B contributes another 1,397 prompt clusters to the Dense-6 core synthesis. Dense-6 paper-facing intervals combine the stored per-family prompt-bootstrap intervals with an independent normal approximation because the 32B raw event records are mirrored separately rather than committed in git. The content/reasoning extension is the harder case: 5,889 event records reduce to 2,983 prompt clusters, and the manuscript reports the cluster-corrected intervals.

Audit levels. We expose the results at three levels, so a reviewer can check the main numbers without trusting prose summaries.

Level	Command	What it verifies	Expected cost
Summary audit	<code>bash scripts/reproduce/reproduce_claims_from_summaries.sh</code>	Recomputes the primary paper numbers from committed JSON/CSV summaries.	CPU only; under 1 minute.
Minimal raw shard	<code>bash scripts/reproduce/reproduce_minimal.sh</code> after fetching the shard	Validates a 20-prompt, one-family raw shard containing cached per-layer logits, intervention outputs, first-divergence records, and expected summary JSONs.	CPU-only if cached logits are used; about 1-3 GPU-hours on one 80GB A100/H100 if regenerated.
Full rerun	Experiment launchers under <code>scripts/run/</code> and source packages under <code>src/poc/</code>	Regenerates full traces, summaries, and plots.	Multi-GPU; see estimates below.

Checkpoint and tokenizer manifest. The current artifact pins Hugging Face model and tokenizer loading through immutable registry revisions in `src/poc/cross_model/config.py`; the shared loader passes those revisions to both `AutoTokenizer.from_pretrained(..., revision=SHA)` and `AutoModelForCausalLM.from_pretrained(..., revision=SHA)`. The manifest generated by `scripts/analysis/build_model_checkpoint_manifest.py` is committed at `results/paper_synthesis/model_checkpoint_manifest.json` and records, for each model, the repo ID, configured immutable revision, current Hub main resolution, tokenizer metadata, and chat-template hash when the tokenizer is accessible; all configured revisions in the manifest equal the audited current main SHA. The reported full-run artifacts were generated before this registry-wide pinning patch and their per-run configs stored repo IDs rather than resolved hashes, so they should be read as repo-ID historical runs whose closest recoverable and now-pinned rerun target is the audited revision below. For Qwen, Mistral, OLMo, and DeepSeek, local Hugging Face cache `refs/main` matched these audited SHAs; Gemma and Llama were gated in the local audit but resolved through the Hub metadata API.

Family	PT repo and pinned/audited SHA	IT repo and pinned/audited SHA	Tokenizer and prompt-template note				
Gemma 3 4B	google/gemma-3-4b-pt cc012e0a6d0787b4adcc0fa2c4da74402494554d	google/gemma-3-4b-it 093f9f388b31de276ce2de164bdc2081324b9767	Gated tokenizer in this audit; code uses raw PT prompts and IT `apply_chat_template(..., ...				
Llama 3.1 8B	meta-llama/Llama-3.1-8B d04e592bb4f6aa9cfee91e2e20afa771667e1d4b	meta-llama/Llama-3.1-8B-Instruct 0e9e39f249a16976918f6564b8830bc894c89659	Gated tokenizer in this audit; code uses raw PT prompts and IT `apply_chat_template(..., ...	eot_id	>` as an additional IT stop token.		
Qwen 3 4B	Qwen/Qwen3-4B-Base 906bfd4b4dc7f14ee4320094d8b41684abff8539	Qwen/Qwen3-4B-1cfa9a7208912126459214e8b04321603b3df60c	Qwen2TokenizerFast, shared vocab size 151669; base EOS `<	endoftext	> (151643), IT EOS <	im_end	> (151645); IT native prompts call `apply_chat_template` with `enable_thinking=False`.
Qwen2.5 32B	Qwen/Qwen2.5-32B 1818d35814b8319459f4bd55ed1ac8709630f003	Qwen/Qwen2.5-32B-Instruct 5ede1c97bbab6ce5cda5812749b4c0bdf79b18dd	Sixth Dense-6 core pair; Qwen2TokenizerFast, shared tokenizer family, raw-shared first-divergence runs ...				
Mistral 7B v0.3	mistralai/Mistral-7B-v0.3 caa1feb0e54d415e2df31207e5f4e273e33509b1	mistralai/Mistral-7B-Instruct-v0.3 c170c708c41dac9275d15a8fff4eca08d52bab71	LlamaTokenizerFast, shared vocab size 32768, shared EOS </s> (2); IT native prompts use the ...				
OLMo 2 7B	allenai/OLMo-2-1124-7B 7df9a82518afdecae4e8c026b27adccc8c1f0032	allenai/OLMo-2-1124-7B-Instruct 470b1fba1ae01581f270116362ee4aa1b97f4c84	GPT2TokenizerFast, shared vocab size 100278, shared EOS `<	endoftext	> (100257) and pad <	pad	> (100277); IT native prompts use the OLMo chat template.
OLMo 2 7B stage progression	Base allenai/OLMo-2-1124-7B 7df9a82518afdecae4e8c026b27adccc8c1f0032	SFT allenai/OLMo-2-1124-7B-SFT 1de02c0175118a9de5854aec80a1f970e701e928; DPO ...	Non-pooled local-transition case study; GPT2TokenizerFast, shared vocab size 100278, no raw prompt ...				
DeepSeek-V2-Lite side case	deepseek-ai/DeepSeek-V2-Lite 604d5664ddddd88a0433dbae533b7fe9472482de0	deepseek-ai/DeepSeek-V2-Lite-Chat 85864749cd611b4353ce1decdb286193298f64c7	MoE appendix side case only; LlamaTokenizerFast, shared vocab size 100002, shared EOS (100001), shared ...				

Across native runs, PT branches use raw prompt text and IT branches use the checkpoint's own chat template through `tokenizer.apply_chat_template([{"role": "user", "content": prompt}], tokenize=False, add_generation_prompt=True)`. Across `raw_shared` first-divergence and residual-state runs, both PT and IT branches are forced to raw text and the code validates identical raw prompt token IDs before comparing residual states.

Claim to command to expected number. The first command below is the most important reviewer audit path: it checks the manuscript numbers directly against committed artifacts.

Claim or number	Command	Expected artifact	Expected number
Dense-5 final-half convergence gap	<code>bash scripts/reproduce/reproduce_claims_from_summaries.sh</code>	<code>results/exp09_cross_model_observational_replication/data/convergence_gap_values.json</code>	tuned 0.410; raw 0.771; raw excluding Gemma 0.712
Endpoint-matched late convergence gap	same	<code>results/paper_synthesis/exp22_endpoint_deconfounded_table.csv</code>	raw +0.425 [+0.356, +0.493]; tuned +0.762 [+0.709, +0.814]; remaining adjacent JS +0.052 [+0.048, +0.057]; future flips +0.203 [+0.190, +0.215]
Endpoint-free matched-prefix JS	same	<code>results/exp16_matched_prefix_js_gap/exp16_js_replay_runpod_20260422_075307/js_summary.json</code>	prompt-mean JS(A', C)=0.121 [0.119, 0.123] pre-corrective and 0.196 [0.193, 0.198] in the final 20%; layer-weighted curve point estimates 0.106 and 0.169

Claim or number	Command	Expected artifact	Expected number
Matched-prefix depth ablation	same	results/exp11_matched_prefix_mlp_graft/plots/exp11_exp3_600rand_v11_depthablation_full/depth_ablation_metrics.json	family-bootstrap intervals over dense-family means: early -0.035 [-0.096, +0.019], middle -0.045 [-0.114, +0.023], late +0.341 [+0.181, +0.502] nats
Symmetric late graft/swap	same	results/exp14_symmetric_matched_prefix_causality/exp13exp14_full_20260416/exp13_full_summary.json	family-bootstrap intervals over dense-family means: PT-side late graft +0.338 [+0.182, +0.504]; IT-side late PT-swap -0.509 [-0.828, -0.224] nats
Late random-control specificity	same	results/exp19_late_mlp_specificity_controls/exp19B_core120_h100x8_20260424_050421_analysis/exp19B_summary_light.json	true +0.327 [+0.298, +0.359]; matched random residual-projection +0.003 [-0.002, +0.008]; true-minus-random +0.324 [+0.294, +0.358] nats
First-divergence identity split	same	results/exp20_divergence_token_counterfactual/factorial_validation_holdout_fast_20260425_2009_with_early/validation_analysis/summary.json	raw-shared PT+IT mid vs PT+IT late: 26.0% [24.5, 27.7] vs 17.6% [16.2, 18.9]; mirror 31.2% [29.6, 32.9] vs 20.8% [19.4, 22.3]
Native late readout loss	same	same	pure IT minus late PT-swap margin drop 13.25 logits [12.91, 13.61]
MLP write-out proxy	same	results/exp21_productive_opposition/exp21_full_productive_opposition_clean_20260426_053736/analysis/summary.json; analysis/effects.csv	late IT-token support +0.789 [+0.754, +0.825]; MLP-only PT-host late graft +0.0035 [-0.001, +0.009]
Residual-state x late-stack context gating (Dense-6 core)	uv run python scripts/analysis/build_exp23_dense6_core_synthesis.py	results/paper_synthesis/exp23_dense6_core/exp23_dense6_core_effects.csv; exp23_dense6_family_effects.csv; exp23_dense6_position_sensitivity.csv; exp23_dense6_interaction.png	common-IT late given PT upstream +0.639 [+0.570, +0.709]; late given IT upstream +3.076 [+2.978, +3.174]; upstream +3.874 [+3.761, +3.987]; late-stack +1.858 [+1.786, ...]
First-divergence position stratification (Dense-6 core)	uv run python scripts/analysis/build_exp23_dense6_core_synthesis.py; five-family raw details via python scripts/analysis/analyze_first_divergence_position_sensitivity.py --n-bootstrap ...	results/paper_synthesis/exp23_dense6_core/exp23_dense6_position_sensitivity.csv; results/paper_synthesis/exp23_position_sensitivity_table.csv; ...	drop position 0 interaction +2.08 [+1.96, +2.19]; position >=3 interaction +1.43 [+1.30, +1.57], with all six family intervals above zero; position >=5 interaction +1.48 ...
First-divergence >=3 category mix	python scripts/analysis/analyze_exp23_position_category_mix.py	results/paper_synthesis/exp23_position_category_mix.csv; results/paper_synthesis/exp23_position_category_mix_note.md	800 records across all five families; prompt mix GOV-CONV 87.5%, GOV-FORMAT 7.6%, SAFETY 4.9%; IT-token mix CONTENT 59.5%, FUNCTION_OTHER 24.9%, FORMAT 15.6%
First-divergence position x prompt category	python scripts/analysis/analyze_exp23_position_prompt_category.py	results/paper_synthesis/exp23_position_prompt_category_effects.csv; results/paper_synthesis/exp23_position_prompt_category_note.md	GOV-CONV interaction all positions +2.05, position >=3 +1.51; GOV-FORMAT position >=3 +2.28; SAFETY position >=3 +0.64 with only 39 clusters
Exp23 label-swap interaction control	uv run python scripts/analysis/analyze_exp23_compatibility_permutation.py --run-root results/exp23_midlate_interaction_suite/exp23_dense5_full_h100x8_20260426_sh4_rw4 --n-permutations ...	results/exp23_midlate_interaction_suite/exp23_dense5_full_h100x8_20260426_sh4_rw4/analysis/compatibility_permutation/	primary IT-late boost +5.56, PT-late boost +2.92, interaction +2.64 over 2,983 prompt clusters; label-swap null 99.9th percentile +0.239, p=5.0e-5
Exp23 subgroup characterization	python scripts/analysis/analyze_exp23_interaction_subgroups.py --run-root results/exp23_midlate_interaction_suite/exp23_dense5_full_h100x8_20260426_sh4_rw4 --n-bootstrap 2000	results/exp23_midlate_interaction_suite/exp23_dense5_full_h100x8_20260426_sh4_rw4/analysis/subgroups/	prompt-category interactions: GOV-CONV +2.05, GOV-FORMAT +3.61, SAFETY +2.83; IT-token-category interactions: CONTENT +2.50, FORMAT +2.60, FUNCTION_OTHER +2.81; assistant- ...
Content/reasoning residual-state extension	bash scripts/run/run_exp23_exp21_content_reasoning_extension.sh	results/exp23_midlate_interaction_suite/exp23_content_reasoning_residual_20260427_0930_h100x8/analysis/exp23_summary.json; analysis/subgroups/exp23_subgroup_report.md	common-IT interaction +1.81 [+1.72, +1.90] over 2,983 prompt clusters from 5,889 event records; prompt interactions: CONTENT-FACT +1.96, CONTENT-REASON +1.35, GOV-FORMAT ...
Content/reasoning label-swap interaction control	uv run python scripts/analysis/analyze_exp23_compatibility_permutation.py --run-root results/exp23_midlate_interaction_suite/exp23_content_reasoning_residual_20260427_0930_h100x8 ...	results/exp23_midlate_interaction_suite/exp23_content_reasoning_residual_20260427_0930_h100x8/analysis/compatibility_permutation/	content/reasoning IT-late boost +4.62, PT-late boost +2.81, interaction +1.81; label-swap null 99.9th percentile +0.178, p=5.0e-5
Content/reasoning MLP write-out extension	same	results/exp21_productive_opposition/exp21_content_reasoning_20260427_0943_h100x8/analysis/summary.json; analysis/effects.csv	late-weight IT-vs-PT MLP margin +0.042 [+0.036, +0.048]; remaining/token-specific component +0.049 [+0.043, +0.055]

Claim or number	Command	Expected artifact	Expected number
Natural-rollout residual-opposition importance	<code>bash scripts/run/run_exp27_natural_rollout_residual_opposition_ntp_runpod.sh; analysis via python scripts/analysis/analyze_exp27_natural_rollout_residual_opposition_ntp.py</code>	<code>results/exp27_natural_rollout_residual_opposition_ntp/exp27_full_dense5_combined_20260430_2050/analysis/exp27_summary.json; exp27_effects.csv; exp27_natural_rollout_ntp.png</code>	remove-opposition NLL hurt: PT +0.0004 [-0.0016, +0.0027], IT +0.0432 [+0.0418, +0.0448], IT-PT +0.0428 [+0.0403, +0.0453]; norm-preserving IT-PT +0.0336 [+0.0312, ...
Qwen2.5-32B Dense-6 family contribution	<code>bash scripts/run/run_exp24_32b_external_validity.sh --model qwen25_32b --phase all; synthesis via 'python scripts/analysis/build_exp24_32b_external_validity_synthesis.py --run-root ...</code>	<code>results/exp24_32b_external_validity/exp24_qwen25_32b_full_eval_v21_20260427_194839/analysis/; results/paper_synthesis/exp24_32b_external_validity/</code>	residual-state interaction +1.446 [+1.321, +1.569]; position >=3 interaction +1.020 [+0.853, +1.205]; PT-upstream late +0.977; IT-upstream late +2.423; raw-KL IT-side ...
OLMo-2 stage progression	<code>RUN_NAME=exp25_olmo_stage_full_20260428_0905 MODE=full bash scripts/run/run_exp25_olmo_stage_progression_runpod.sh; synthesis via 'python ...</code>	<code>results/paper_synthesis/exp25_olmo_stage_full_20260428_0905/olmo_stage_progression_table.csv; olmo_stage_progression_summary.json; olmo_stage_preflight.json; ...</code>	Base→SFT interaction +0.782 [+0.681, +0.882]; SFT→DPO +0.135 [+0.114, +0.157]; DPO→RLVR +0.016 [+0.008, +0.025]; Base→RLVR +1.930 [+1.749, +2.112]
Behavioral sanity check and human audit	same	<code>results/exp15_symmetric_behavioral_causality/plots/exp15_eval_core_600_t512_dense5/exp15_behavior_summary.json; ...</code>	LLM resolved G2: 56.3% [53.3, 59.5] and 77.1% [74.7, 79.6]; human resolved G2: 60.2% [52.9, 67.5] and 73.1% [66.5, 79.5]

Committed versus regenerated artifacts. The repository commits paper-facing summaries and plots used for manuscript numbers: JSON/CSV/MD summaries, bootstrap confidence intervals, human-evaluation summaries, and final PNG figures. It does not commit large regenerated intermediates such as raw activation arrays (* . npy, * . npz), model/probe tensors (* . pt, * . safetensors), tuned-lens checkpoints, or multi-gigabyte raw per-token trace JSONL/GZ files. These are size exclusions, not confidentiality exclusions. The corresponding collection scripts, analysis scripts, prompt manifests, and anonymized archive pointers are public within the review artifact.

Large-artifact locations. During review, raw traces, tuned-lens probes, endpoint-control records, late random-control traces, first-divergence archives, and the minimal audit shard are mirrored under the anonymized artifact archive. The archive contains concrete paths for review; the public camera-ready release will replace anonymized review paths with permanent non-anonymous storage locations after the double-blind period.

Hardware and runtime. All full reruns use bf16 inference and deterministic greedy decoding unless a script states otherwise. The 4B-8B Dense-5 experiments fit on one 80GB A100/H100 per model job; Qwen2.5-32B is a separate multi-GPU Exp24 run that supplies the sixth Dense-6 core family and is costed separately from the older Dense-5 suite. DeepSeek-V2-Lite is reported separately because it is MoE/SFT-only. The full Dense-5 intervention and endpoint-control suite (Exp11/14/16/20/21/22/23, including trace collection and analysis) should be budgeted as hundreds of H100 GPU-hours and roughly 0.5-1.5 TB of transient disk if raw traces and per-token records are retained; reproducing the Dense-6 core first-divergence result additionally requires the Qwen2.5-32B Exp24 run or the committed paper-facing synthesis artifacts. The minimal 20-prompt shard needs one 80GB A100/H100, about 1-3 GPU-hours if regenerated from checkpoints, and about 20-80 GB local disk depending on whether all per-layer logits are kept. The summary audit is CPU-only and reads only committed JSON/CSV files.

Tuned-lens reproducibility. The main intervention claims do not require tuned-lens checkpoints. The matched-prefix JS replay, graft/swap KL summaries, first-divergence identity/margin results, and finite-difference MLP write-out analyses are auditable from committed summaries and do not depend on trained probes. The tuned lens is used for the layerwise visualization and commitment diagnostics, and the paper reports raw-lens sensitivity alongside it. Probe checkpoints are included in the anonymized artifact archive; retraining uses `uv run python -m src.poc.cross_model.tuned_lens --model MODEL --variant {pt,it} --device cuda:N`. A complete dense-5 retrain is 10 PT/IT runs; including the DeepSeek side case is 12 runs. The full set takes about 4-6 wall-clock hours on 8x80GB A100/H100 GPUs with joint all-layer training.

Main claim-to-artifact map.

Claim or analysis	Code endpoints	Committed artifacts
Dense-5 convergence gap and commitment delay, with DeepSeek side case in appendix artifacts	<code>src/poc/exp09_cross_model_observational_replication/</code> ; shared adapters in <code>src/poc/cross_model/</code>	<code>results/exp09_cross_model_observational_replication/data/</code> , <code>results/exp09_cross_model_observational_replication/plots/</code>
Endpoint-matched convergence gap	<code>src/poc/exp22_endpoint_deconfounded_gap/</code> ; <code>scripts/run/run_exp22_endpoint_deconfounded_gap_runpod.sh</code> ; <code>scripts/analysis/analyze_exp22_endpoint_deconfounded_gap.py</code> ; ...	<code>results/paper_synthesis/exp22_endpoint_deconfounded_table.csv</code> , <code>results/paper_synthesis/exp22_endpoint_deconfounded_summary.png</code> , raw mirror listed above
Gemma feature and Tier-0 validation controls	<code>src/poc/exp06_corrective_direction_steering/</code> ; <code>src/poc/exp07_methodology_validation_tier0/</code> ; <code>scripts/run/run_exp7_0*.sh</code> ; <code>scripts/plot/plot_validation_tier0.py</code>	<code>results/exp06_corrective_direction_steering/plots/</code> , <code>results/exp07_methodology_validation_tier0/data/</code> , <code>results/exp07_methodology_validation_tier0/plots/</code>

Claim or analysis	Code endpoints	Committed artifacts
Layer-range and onset sensitivity	scripts/plot/plot_validation_tier0.py; Tier-0 OF/OJ runs	results/exp07_methodology_validation_tier0/0F/layer_range_sensitivity_table.csv, results/exp07_methodology_validation_tier0/0J/onset_table.csv, ...
Matched-prefix graft depth ablation	src/poc/exp11_matched_prefix_mlp_graft/	results/exp11_matched_prefix_mlp_graft/plots/exp11_exp3_600rand_v11_depthablation_full/ and selected summaries under results/exp11_matched_prefix_mlp_graft/data/
Symmetric graft/swap and late random-control specificity	src/poc/exp14_symmetric_matched_prefix_causality/; src/poc/exp19_late_mlp_specificity_controls/	results/exp14_symmetric_matched_prefix_causality/exp13exp14_full_20260416/, results/exp19_late_mlp_specificity_controls/
Behavioral sanity check and human audit	src/poc/exp15_symmetric_behavioral_causality/; scripts/eval/llm_judge.py; human-audit materials under paper_draft/human_eval_survey/	results/exp15_symmetric_behavioral_causality/plots/exp15_eval_core_600_t512_dense5/, results/exp15_symmetric_behavioral_causality/human_eval/
Natural-rollout residual-opposition importance	src/poc/exp27_natural_rollout_residual_opposition_ntp/; Exp27 analysis scripts under scripts/analysis/	results/exp27_natural_rollout_residual_opposition_ntp/exp27_full_dense5_combined_20260430_2050/analysis/
Endpoint-free matched-prefix JS replay	src/poc/exp16_matched_prefix_js_gap/; scripts/analysis/analyze_exp16.py; scripts/plot/plot_exp16.py	results/exp16_matched_prefix_js_gap/exp16_js_replay_runpod_20260422_075307/, results/exp16_matched_prefix_js_gap/exp16_js_reverse_pt_teacher_20260422_165259/
Native final-token flow and matched-prefix chronology	src/poc/exp18_midlate_token_handoff/; scripts/run/run_exp18_yanda_full.sh	results/exp18_midlate_token_handoff/full_runpod_20260423_095122/, results/exp18_midlate_token_handoff/matched_prefix_latest/
First-divergence token identity and margin amplification	src/poc/exp20_divergence_token_counterfactual/; Exp20 analysis scripts under scripts/analysis/	results/exp20_divergence_token_counterfactual/full_runpod_20260423_2148_combined_final/deep_dive/, ...
MLP write-out, local context-gating proxy, content/reasoning extension, and synthesis figure	src/poc/exp21_productive_opposition/; scripts/run/run_exp21_productive_opposition_runpod.sh; scripts/run/run_exp23_exp21_content_reasoning_extension.sh; ...	results/exp21_productive_opposition/exp21_full_productive_opposition_clean_20260426_053736/analysis/, ...
Mid+late KL factorial, residual-state x late-stack interaction, subgroup/position characterization, label-swap control, and content/reasoning extension	src/poc/exp23_midlate_interaction_suite/; src/poc/exp23_midlate_kl_factorial/; scripts/run/run_exp23_midlate_interaction_suite.sh; ...	results/exp23_midlate_interaction_suite/exp23_dense5_full_h100x8_20260426_sh4_rw4/analysis/, ...
Qwen2.5-32B Dense-6 core member	scripts/run/run_exp24_32b_external_validity.sh; scripts/analysis/build_exp24_32b_external_validity_synthesis.py; scripts/analysis/analyze_exp24_32b_position_sensitivity.py; ...	results/exp24_32b_external_validity/exp24_qwen25_32b_full_eval_v21_20260427_194839/analysis/, results/paper_synthesis/exp24_32b_external_validity/, ...
OLMo-2 stage progression	src/poc/exp25_olmo_stage_progression/; scripts/run/run_exp25_olmo_stage_progression_runpod.sh; scripts/analysis/analyze_olmo_stage_progression.py	results/paper_synthesis/exp25_olmo_stage_full_20260428_0905/, results/exp20_divergence_token_counterfactual/exp25_olmo_stage_full_20260428_0905_exp20/, ...